



Faculty of Technology & Engineering
Bachelor of Technology Programme
Electronics and Communication Engineering
B.Tech. (EC)

ACADEMIC
REGULATIONS
&
SYLLABUS
(Choice Based Credit System)

Academic Year: 2023-2024



Chandubhai S. Patel Institute of Technology
V.T. Patel Department of Electronics and Communication Engineering

Vision

“To lead in creating globally competitive professionals ready to take up industrial and societal challenges in Electronics and Communication Engineering field.”

Mission

To provide a creative learning environment for excellence in technical education.

THE PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

- PEO1 To prepare the student(s) for successful career as an engineer, a corporate or a government professional, a scientist, an academician, a technocrat, an administrator and an entrepreneur.
- PEO2 To make students demonstrate their abilities to adapt to a rapidly changing environment by having learning approach and apply new skills and new technologies to solve the problems.
- PEO3 To create an ambience where the students are cared for in every aspect and motivated to become excellent working professionals who will continue to cherish their association with the organization as a whole, staff and colleagues.
- PEO4 To provide continued professional development and lifelong learning throughout their career to inculcate strong teamwork and by installing lacking skills for the benefit of the society.
- PEO5 To prepare the students to apply their technical skill(s) to analyse and design appropriate solution(s) with social consciousness and ethical values.

PROGRAM ARTICULATION MATRIX

(Not Matching, 1-Low, 2,3-High)

Mission Statement	PEO1	PEO2	PEO3	PEO4	PEO5
To provide world-class educational activities	3	3	3	3	3
To deploy latest technologies	-	2	3	3	-
To foster research activities in consultation with academia and industry	3	2	2	3	3

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

- PO1 **Engineering knowledge:** Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2 **Problem analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3 **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
- PO4 **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5 **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
- PO6 **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7 **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8 **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9 **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10 **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11 **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12 **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

FACULTY OF TECHNOLOGY AND ENGINEERING
ACADEMIC REGULATIONS
Bachelor of Technology Programmes
Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. *System of Education*

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. *Duration of Programme*

(i)	Undergraduate programme	(B. Tech.)
	Minimum	8 semesters (4 academic years)
	Maximum	16 semesters (8 academic years)

3. *Eligibility for admissions*

As enacted by Govt. of Gujarat from time to time.

4. *Mode of admissions*

As enacted by Govt. of Gujarat from time to time.

5. *Programme structure and Credits*

As per annexure – 1 attached

6. *Attendance*

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be Minimum 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

- 7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
- 7.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

7.2 University Examination

- 7.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper and 100% for practical test or oral test or presentation by the student or a combination of any two or more of these.
- 7.2.2 **In order to earn the credit in a course a student has to obtain grade other than FF.**

7.3 Performance at Internal & University Examination

- 7.3.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per subject	Minimum marks Overall per subject
40%	45%

- 7.3.2 A student failing to score 45% of the final examination will get a FF grade.
- 7.3.3 If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	≥73 <80	≥66 <73	≥60 <66	≥55 <60	≥50 <55	≥45 <50	<45
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i)
$$SGPA = \frac{\sum C_i G_i}{\sum C_i}$$
 where C_i is the number of credits of course i
 G_i is the Grade Point for the course i

and $i = 1$ to n , n = number of courses in the semester

- (ii) $CGPA = \sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.
- (iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.
- (iv) A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- (v) A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

9. Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

- 9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and
- 9.1.2 He should have cleared all internal and external evaluation components in every course; and
- 9.1.3 He should have secured a minimum CGPA of 4.5 at the end of the programme; and
- 9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.

9.2 The student who fails to satisfy minimum requirement of CGPA at the end of program will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	$CGPA \geq 7.5 \ \& \ \leq 10.0$
First class:	$CGPA \geq 6.0 \ \& \ < 7.5$
Second Class:	$CGPA \geq 5.0 \ \& \ < 6.0$
Pass Class:	$CGPA < 5.0$

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Choice Based Credit System

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

1.1. Core Courses

1.1.1 University Core (UC)

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/specialisation.

1.1.2 Programme Core (PC)

A ‘Core Course’ is a course which acts as a fundamental or conceptual base for Chosen Specialisation of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

1.2 Elective Course (EC)

An ‘Elective Course’ is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the University/Institute/Department (as the case may be) from time to time.

1.2.1 Institute Elective Course (IE)

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

1.2.2 Programme Elective Course (PE)

A ‘Programme Elective Course’ is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time

1.2.3 Cluster Elective Course (CE)

An ‘Elective Course’ is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.3 Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student's Grade Sheet. Attendance and Course Assessment is compulsory for Non Credit Courses

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN EC ENGINEERING											
CHOICE BASED CREDIT SYSTEM											
Sem.	Course Code	Course title	Teaching Scheme				Examination Scheme				Total
			Contact Hrs.			Credits	Theory		Practical		
			Theory	Practical	Total		Internal	External	Internal	External	
SEM 3	MA252	Advance Engineering Mathematics	4	0	4	4	30	70	0	0	100
	EC260	Design of Digital Circuits	4	2	6	5	30	70	25	25	150
	EC261	Circuit Theory	2	2	3	3	30	70	25	25	150
	EC253	Electronic Devices & Measurement	4	2	6	5	30	70	25	25	150
	HS121.01A	Creativity, Problem Solving and Innovation	0	2	2	2	0	0	30	70	100
	EC281.01	Introduction to MATLAB Programming (University Elective)	0	2	2	2	30	70	0	0	100
			14	10	23	21	150	350	105	145	750
SEM 4	EC263	Control Theory	3	0	3	3	30	70	0	0	100
	EC256	Analog Circuits & Applications	4	2	6	5	30	70	25	25	150
	EC257	Microprocessor & Interfacing	3	2	5	4	30	70	25	25	150
	EC262	Python Programming	0	4	4	4	0	0	50	50	100
	HS111.02A	Human Values and Professional Ethics	0	2	2	2	30	70	0	0	100
	EC282.01	Prototyping Electronics with Arduino (University Elective)	0	2	2	2	30	70	0	0	100
	EC296	Machine Learning	0	4	4	4	0	0	50	50	100
			10	16	26	24	150	350	150	150	800

Note:

- **University Elective (UE):-** University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects like Research Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc. can be included.
- **Cluster Elective (CT):-** Institutional Electives means common electives among a cluster of programmes (eg. CE/IT/EC/EE etc.). If Institutional Electives are not applicable, it will be Programme electives
- **Programme Elective (PE):-**
- **Institute Elective (IE):-**
- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/program(s)

Elective Course:

Subject Code	Subject Name
EC271	Hardware Modelling using Verilog
EC272	Introduction to Robotics
EC273	Fundamentals of Artificial Intelligence
EC274	Ethics in Engineering Practice
EC275	Project Management

University Elective - I (UE - I)	University Elective - II (UE - II)
Introduction to MATLAB Programming	Prototyping Electronics with Arduino
Art of Programming	Web Designing
Environmental Sustainability and Climate Change	Basics of Environmental Impact Assessment
Python for Programming	Computer Programming for Electrical Engineering
WEB DESIGNING & UI/UX	Data Visualization
Engineering Drawing	Material Science
Fundamentals of Packaging	Cosmetics in daily life
Basic Laboratory Techniques	Life Style Diseases & Management
First Aid & Life Support	Occupational Health & Ergonomics
Health Promotion and Fitness	Programming the Internet
Introduction to Web Designing	Health Care Management
Banking and Insurance	Astrophysics, Space And Cosmos-II
Astrophysics, Space And Cosmos-I	Maintenance Of Household Apparatus
Sdg Handprint Laboratory	MATLAB Programming

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)											
TEACHING & EXAMINATION SCHEME FOR B TECH IN ELECTRONICS & COMMUNICATION ENGINEERING											
CHOICE BASED CREDIT SYSTEM											
Sem	Course Code	Course title	Teaching Scheme				Examination Scheme				Total
			Contact Hrs.			Credits	Theory		Practical		
			Theory	Practical	Total		Internal	External	Internal	External	
SEM 5	EC354	Electromagnetics & Antenna Technology	4	2	6	5	30	70	0	0	100
	EC323	VLSI Technology	4	0	4	4	30	70	0	0	100
	EC357	Digital Signal Processing	3	2	5	4	30	70	25	25	150
	EC358	Microcontroller & Application	3	2	5	4	30	70	25	25	150
	ECXXX	Programme Elective-I	0	2	2	2	30	70	25	25	150
	EC360	Summer Internship-I	0	3	3	3	0	0	75	75	150
	HS124.01A	Professional Communication	0	2	2	2	0	0	30	70	100
	EC314	Modeling using Verilog	0	4	4	4	0	0	50	50	100
	EC396	Elements of Internet of Things	0	4	4	4	0	0	50	50	100
		14	21	35	32	150	350	280	320	1100	
SEM 6	EC319	Design, Testing and Verification	4	2	6	5	30	70	25	25	150
	EC362	Digital Communication	3	2	5	4	30	70	25	25	150
	EC368	Introduction to Embedded Linux	0	2	2	2	0	0	25	25	50
	EC364	Embedded Systems	3	2	5	4	30	70	25	25	150
	ECXXX	Programme Elective - II	0	2	2	1	30	70	25	25	150
	HS132.02A	Contributor Personality Development	2		2	2	25	25	25	25	100
	EC322	System Verilog Programming	0	4	4	4	0	0	50	50	100
	EC397	Elements of Robotics	0	4	4	4	0	0	50	50	100
			12	18	30	26	145	305	250	250	950

Note:

- **University Elective (UE):** - University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects like Research Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc. can be included.
- **Cluster Elective (CT):** - Institutional Electives means common electives among a cluster of programmes (eg. EC/CE/IT/EE etc.). If Institutional Electives are not applicable, it will be Programme electives
- **Programme Elective (PE):** -
- **Institute Elective (IE):** -
- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/program(s)

ELECTIVES	Code	Programme Elective - I	Code	Programme Elective - II
	EC385	Mobile Application Development	EC321	Programming in JAVA
	EC318	Simulation on RF and Microwave Engineering	EC320	Artificial Intelligence

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)											
TEACHING & EXAMINATION SCHEME FOR B TECH IN ELECTRONICS & COMMUNICATION ENGINEERING											
CHOICE BASED CREDIT SYSTEM											
Sem	Course Code	Course title	Teaching Scheme				Examination Scheme				Total
			Contact Hrs.			Credits	Theory		Practical		
			Theory	Practical	Total		Internal	External	Internal	External	
SEM 7	EC451	Data Communication & Networking	4	2	6	5	30	70	25	25	150
	EC452	RF & Microwave Engineering	3	2	5	4	30	70	25	25	150
	EC453	Wireless Communication	3	2	5	4	30	70	25	25	150
	EC484	Fiber Optic Communication Technology	3	0	3	3	30	70	0	0	100
	ECXXX	Programme Elective - III	3	2	5	4	30	70	25	25	150
	EC486	Minor Project	0	2	2	1	0	0	25	25	100
	EC456	Summer Internship-II	0	3	0	3	0	0	75	75	150
	EC496	Sensor Networks	4	2	6	5	30	70	25	25	150
			20	15	32	29	180	420	225	225	1100
SEM 8	EC 458	Project	0	36	36	18	0	0	250	350	600
			0	36	36	18	0	0	250	350	600

Note:

- **University Elective (UE):** - University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects like Research Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc. can be included.
- **Cluster Elective (CT):** - Institutional Electives means common electives among a cluster of programmes (eg. CE/IT/EC/EE etc.). If Institutional Electives are not applicable, it will be Programme electives
- **Programme Elective (PE):** -
- **Institute Elective (IE):** -
- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/program(s)

List of Programme Elective -III	
EC481	Digital Image Processing
EC482	Radar systems
EC483	Error Control Coding

**B. Tech. (Electronics and Communication Engineering)
Programme**

SYLLABI
(Semester – 3)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF APPLIED SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES
MA252: ADVANCE ENGINEERING MATHEMATICS
B TECH 3rd SEMESTER (EC ENGINEERING)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	-	4	4
Marks	100	-	100	

A. Objective of the Course:

This course is an important course to understand courses like Electromagnetism, Digital Communication, Digital Signal Processing, Control Theory of Electronics and Communication Engineering/ Electrical Engineering.

The objectives of the course are:

1. to understand the concepts of Fourier coefficients and Fourier series, understand the concepts of Laplace transforms to solve differential equations,
2. to understand basic concepts like linear dependence and independence,
3. to understand differentiation of vector fields, determining gradient, curl and directional derivatives with physical interpretations,
4. to understand the concepts of Z– transform of a sequence and derive transforms of specified sequences,
5. to understand concept of Fourier Transform for analysing aperiodic and periodic signals.

B. Outline of the course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Laplace Transforms and Its Applications	15
2.	Fourier Series	10
3.	Linear Algebra	07
4.	Vector Differential Calculus	08
5.	Z–transforms	10
6.	Fourier Transforms	10

Total hours: 60

C. Detailed Syllabus:

1.	Laplace Transforms and Its Applications:	15 Hours 25%
1.1	Laplace transforms as an improper integral and its existence.	
1.2	Laplace transforms of elementary functions and its properties.	
1.3	Inverse Laplace transforms and its properties.	
1.4	First and second shifting theorems. Laplace transforms of derivatives and integrals.	
1.5	Convolution theorem and its application to obtain inverse Laplace transform.	
1.6	Laplace transform of periodic functions, Unit step function, Unit impulse function (Dirac delta function).	
1.7	Solving differential equations using Laplace transform.	
1.8	Applications of Laplace transform to ODE: Electric circuits (RLC and RL)	
2.	Fourier Series:	10 Hours 17%
2.1	Periodic functions, Dirichlet's conditions, Trigonometric series.	
2.2	Euler formulae, Fourier series of periodic functions of period p.	
2.3	Fourier series of Discontinuous functions.	
2.4	Fourier series of Even and odd functions; half range Fourier series.	
3.	Linear Algebra:	07Hours 11%
3.1	Introduction of Vector Space and its examples.	
3.2	Linear dependence and independence and its examples.	
3.3	Basis and its examples.	
4.	Vector Differential Calculus:	08 Hours 13%
4.1	Revision of concepts of Vector algebra, Scalar and Vector fields.	
4.2	Gradient of a scalar functions, Directional derivatives.	
4.3	Divergence and Curl of a vector field and their properties.	
4.4	Physical interpretations of gradient, divergence and curl: Irrotational and solenoidal vector fields.	
4.5	Scalar potential function.	
5.	Z-transforms:	10 Hours 17%

5.1	Introduction to Z– transform. Representation of sequences.	
5.2	Z–transforms of basic sequences.	
5.3	Unit impulse and Unit step sequence. Basic operations on sequence.	
5.4	Properties of Z–transforms: Damping rule and shifting property.	
5.5	Introduction to inverse Z–transforms. Inverse Z – transform using partial fraction and geometric series.	
6.	Fourier Transforms:	10 Hours 17%
6.1	Introduction to Fourier integral: sine and cosine integrals.	
6.2	Introduction to Fourier Transform and inverse Fourier Transform.	
6.3	Properties and examples of Fourier transforms.	
6.4	Fourier sine and cosine transforms.	

D. Instructional Method and Pedagogy:

- At the starting of the course, the course delivery pattern, prerequisite of the subject must be discussed.
- Lectures may be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 5% component of the overall evaluation.
- Minimum two internal tests/ unit tests will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Quizzes (surprise tests) /Oral tests/ Viva/Assignment will be conducted which carries 10% component of the overall evaluation.

E. Student Learning Outcomes:

- At the end of the course, the students would able to know the various applications of engineering Mathematics in their respective field, e.g. Vector calculus is mainly useful for Electromagnetism. Fourier series, Laplace transforms, Z-transforms and applications of differential equations are widely applicable in Control Theory, Communication Engineering and Signal Processing etc.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Analyze the Fourier series of periodic functions and determine the solution of complex engineering problems. Solutions of heat, wave and Laplace's equations using it.
CO2	Classify Laplace transforms and inverse Laplace transforms and its techniques to solve ordinary differential equations of real world problems.
CO3	Z transform and inverse Z transforms of elementary sequences and use it to solve compound engineering problems.

CO4	Fundamental mathematical properties of the Fourier transform and inverse Fourier transform including linearity, shift, symmetry, scaling, modulation and convolution.
CO5	Dot product, cross product, length of vectors, partial derivatives, derivatives of vector-valued functions, gradient functions and these tools are employed successfully in different branches of engineering.
CO6	Infer important concepts of linear algebra and apply them to solve problems in the engineering domain related to Linear Algebra.

• Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	1	-	1	-	-	1	-	1	2	2
CO2	3	2	1	1	1	-	1	-	-	1	-	1	2	2
CO3	3	2	-	1	1	-	-	-	-	1	-	1	2	2
CO4	3	2	-	1	1	-	-	-	-	1	-	1	2	2
CO5	3	2	2	1	1	-	1	-	-	1	-	1	2	2
CO6	2	2	1	-	-	-	-	-	-	1	-	1	1	1

- Correlation levels 1, 2 or 3 as defined below:
- 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. Recommended Study Material:

❖ Text Books:

1. Erwin Kreyszig; Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
2. Debnath, Lokenath, and Dambaru Bhatta; Integral transforms and their applications. CRC press, 2014.
3. K. A. Stroud and D. J. Booth; Advanced Engineering Mathematics, Palgrave Macmillan, 2006.

❖ Reference Books:

1. Howard Anton and Chris Rorres; Elementary linear algebra. John Wiley & Sons, 2010.
2. H. K. Dass; Advanced engineering mathematics, S. Chand, 2008.
3. M. D. Greenberg; Advanced Engineering Mathematics, 2nd ed., Pearson, 1998.
4. B. S. Grewal; Higher engineering mathematics. 43rd Ed., Khanna Publishers, New Delhi, 2014.

5. M. D. Weir et. al.; Thomas' Calculus, 11th Ed., Pearson Education, 2008.
6. C. Ray Wylie and L. C. Barrett; Advanced Engineering Mathematics, McGraw Hill pub., 1982.

❖ **URL link:**

1. <http://nptel.ac.in/courses/111107098/>
2. <http://www1.maths.leeds.ac.uk/~frank/math2420/notes.pdf>

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC260: DESIGN OF DIGITAL CIRCUITS

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction	02
2.	Number System & Binary Codes	04
3.	Logic Gates & Boolean Algebra	04
4.	Minimization of Switching Functions	10
5.	Combinational Logic Design	12
6.	Programmable Logic Devices	05
7.	Flip Flops	10
8.	Shift Registers	05
9.	Counters	08
	Total Hours (Theory):	60
	Total Hours (Lab):	30
	Total Hours:	90

Detailed Syllabus:

- | | | | |
|-----------|--|----------------|-----------|
| 1. | Introduction | 2 Hours | 3% |
| 1.1 | Fundamentals of Digital & Analog Systems | | |
| 1.2 | Digital Logic Levels | | |

1.3	Digital Integrated Circuits, Level of Integration		
1.4	Digital IC Specification Terminologies		
2.	Number System & Binary Codes	4 Hours	6%
2.1	Binary Number System		
2.2	Representation of Signed Numbers and Binary Arithmetic		
2.3	Octal Number System		
2.4	Hexadecimal Number System		
2.5	Classification of Binary Codes		
2.6	Natural BCD Code		
2.7	Excess-3 Code		
2.8	Gray Code		
3.	Logic Gates & Boolean Algebra	4 Hours	9%
3.1	Introduction of AND, OR & NOT Gate, Universal Gates, EX-OR & EX-NOR Gates and associated Logic Operations		
3.2	Axioms & Laws of Boolean Algebra		
3.3	Duality Theorems		
3.4	Reducing Boolean Function		
3.5	AND-OR-INVERT Implementation of Boolean Expression		
4	Minimization of Switching Functions	10 Hours	16%
4.1	Expansion of Boolean Function into SOP & POS Forms		
4.2	Implementation of Logic Expression using Universal Gates		
4.3	Minimization of Boolean Expression using K-Map (Up to 5 Variable K-Map)		
4.4	Prime Implicants & its types		
4.5	K-Map with Don't Care Combination, Limitation of K-map		
5	Combinational Logic Design	12 Hours	20%
5.1	Half Adder & Full Adder, Half Subtractor & Full Subtractor		
5.2	Binary Parallel Adder		
5.3	Look ahead Carry Adder		
5.4	Code Converters		
5.5	Parity Bit Generator / Checkers		
5.6	Comparators		
5.7	Encoder & Priority Encoders		

5.8	Decoder		
5.9	Multiplexers & Demultiplexer		
5.10	Modular Design Using Chips		
5.11	Hazard & Hazard Free Realization		
6	Programmable Logic Devices	5 Hours	8%
6.1	Design of ROM		
6.2	Design of PAL & PLA Circuits		
6.3	Introduction of FPGA & CPLD		
7	Flip-Flops	10 Hours	17%
7.1	Introduction of Sequential Circuit		
7.2	Latches & Flip-Flop		
7.3	Asynchronous Inputs		
7.4	Flip Flop Operating Characteristics		
7.5	Clock Skew & Time Race		
7.6	Race Around Conditions & Master Slave Flip-Flop		
7.7	Introduction to State Model		
7.8	Flip Flop Excitation Table		
7.9	Conversion of Flip-Flop		
8	Shift Registers	5 Hours	8%
8.1	Buffer Register & Controlled Buffer Register		
8.2	Types of Shift Register		
8.3	Universal Shift Register		
9	Counters	8 Hours	13%
9.1	Difference between Synchronouns & Asynchronous Counter		
9.2	Ripple Counter		
9.3	Design of Asynchronous Counter		
9.4	Design of Synchronous Counter		

Course Outcomes (CO)

At the end of the course student will be able to:

CO1	Selection of proper digital electronics techniques or systems for required applications.
CO2	Understand the need for digital electronics for various applications
CO3	Understand the concept of chip designing and IC fabrication with real time examples.
CO4	Evaluate various logic circuit using gates registers, counters, flip-flops, controllers and processors with latest tools.
CO5	Design logic circuits at individual levels as well as in team form.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3	-	-	-	2	-	-	-
CO2	3	3	3	3	3	-	-	-	-	-	-	-
CO3	3	3	3	3	3	-	-	-	2	-	2	1
CO4	3	3	3	3	3	-	-	-	3	-	-	2
CO5	2	2	2	2	2	-	-	-	2	2	-	

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Digital Principles and Application, Anand Kumar, 4th Edition, PHI Publication
2. Digital Logic and Computer Design, M Morris Mano, PHI Publication
3. Digital System Design Using VHDL, Charles H. Roth, Cengage Publication

❖ Web Materials/ Reading Material:

1. <http://zebu.uoregon.edu/~rayfrey/432/DigitalNotes.pdf>
2. <http://smendes.com/el10b/gates1.gifz>
3. <https://www.tinkercad.com/>
4. Hand Outs, Assignments, Question Banks

❖ Simulators:

1. Xilinx Altera

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC261: CIRCUIT THEORY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	2	3	3
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Basic of Concept and Network Representation	05
2.	Nodal Analysis and Mesh Analysis of resistive Circuits	08
3.	Initial Conditions	08
4.	Time domain response of First order RL & RC circuits and Time domain response of Second order circuits	08
5.	Network Analysis Using Laplace Transform	08
6.	Two Port Networks	07
7.	Network Theorems	08
8.	Network Topology	08
	Total Hours (Theory):	60
	Total Hours (Lab):	30
	Total Hours:	90

Detailed Syllabus:

- 1. Basic Of Concept And Network Representation. 5 Hours 09%**
 - 1.1 Electromotive Force, Potential and Voltage , A Voltage Source with a Resistance Connected at its Terminals, Two terminal Capacitance

	Two terminal Inductance, Ideal Independent Two terminal Electrical Sources, Classification of Two terminal Elements.		
1.2	Dot Convention.		
2.	Nodal Analysis and Mesh Analysis of resistive Circuits.	8 Hours	13%
2.1	Nodal Analysis of Circuits Containing Resistors and Independent Sources ,Nodal Analysis of Circuits Containing Dependent Voltage Sources		
2.2	Mesh Analysis of Circuits with Independent Sources, Mesh Analysis of Circuits Containing Dependent Sources		
2.3	Source Transformation Theorem for circuits with independent sources , Source Transformation Theorem for circuits with Dependent sources		
3.	Initial Conditions.	8 Hours	13%
3.1	Initial conditions in elements, procedure for evaluating initial conditions, Solution of circuit equations by using Initial Conditions.		
4.	Time domain response of First order RL & RC circuits and Time domain response of Second order circuits.	8 Hours	13 %
4.1	Mathematical preliminaries, Source free response, DC response of first order circuits.		
4.2	Source free second order linear networks, second order linear networks with constant inputs.		
5.	Network Analysis Using Laplace Transform.	8 Hours	13%
5.1	Laplace Transformation, Inverse Laplace Transformation, Partial Fraction Expansion, Applications to Circuit Analysis.		
5.2	Waveform Synthesis, Laplace Transform of periodic functions.		
6.	Two Port Networks.	7 Hours	13%
6.1	Open circuit Impedance Parameters.		
6.2	Short Circuit Admittance Parameters, Transmission Parameters.		
6.3	The Hybrid Parameters, Relationships Between Parameters Sets.		
7.	Network Theorems.	8 Hours	13%
7.1	Super position Theorem.		

7.2 Thevenin's Theorem, Norton's Theorem.

7.3 Maximum power transfer theorem, Reciprocity theorem, and Millman's theorem.

8. Network Topology.

8 Hours

13%

8.1 Incidence matrix, Loop matrix.

8.2 Cut sets and Cut set matrix, Examples.

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Remember transient response, Steady state response, network function.
CO2	Understand the knowledge of basic circuital law and simplify the network using reduction techniques.
CO3	Understand tie set and cut set methods for solving various circuits.
CO4	Analyze the circuit using Kirchhoff's law and Network simplification theorems.
CO5	Evaluate two-port network parameters, design attenuators and equalizers.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	3	-	-	-	-	2	-	-	1
CO2	3	3	3	3	-	-	-	-	2	-	-	1
CO3	2	3	3	3	-	-	-	-	2	-	-	1
CO4	2	3	3	1	-	-	-	-	2	-	-	1
CO5	2	3	3	2	-	-	1	-	2	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ **Reference Books:**

1. Network Analysis By Van Valkenburg, 3rd Edition, McGraw-Hill Science/Engineering/Math
2. Circuit and Networks by Sudhakar McGraw-Hill Science/Engineering/Math
3. Network Analysis and Synthesis By G.K. Mithal
4. Electrical Circuit Analysis by C.L. Vadhwa, New Age International

❖ **Web Materials/ Reading Material:**

1. www.educyclopedia.be/electronics/electricitycircuits.htm
2. www.ebookchm.com/ebook___network-analysis-in-circuit-theory-ppt_.html
3. Hand Outs
4. Assignments
5. Question Bank

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC253: ELECTRONIC DEVICES & MEASUREMENT

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Semiconductor Physics	07
2.	Transistor Characteristics and Transistors Biasing and Thermal	08
3.	Field Effect Transistor	08
4.	Multistage Amplifiers	07
5.	Measurement Errors	03
6.	Bridge Measurements	05
7.	Digital Instruments	08
8.	Digital Storage Oscilloscope	04
9.	Function Generator And Analyzer	04
10.	Data Acquisition Systems And Transducers	06
	Total Hours (Theory):	60
	Total Hours (Lab):	30
	Total Hours:	90

Detailed Syllabus:

- 1. Semiconductor Physics** **7 Hours 10%**
 - 1.1 The Energy Band Theory of Crystals, Insulators, Semiconductors and Metals, Mobility and Conductivity, Electrons and Holes In An Intrinsic

1.2	Semiconductor, Donor and Acceptor Impurities, Charge Densities Mobility and Conductivity, Electrons and Holes in an Intrinsic Semiconductor, Donor and Acceptor Impurities, Charge Densities		
2.	Transistor Characteristics and Transistors Biasing and Thermal Stabilization	8 Hours	12%
2.1	Junction Transistor, Transistor Action, Transistor as a Amplifier, Transistor Configurations- CB, CC, CE, CE Cut-off, And Saturation Regions		
2.2	Bias Stabilization, Stabilization Techniques, Bias Compensation, Thermal Runaway, Phototransistor		
3.	Field Effect Transistor	8 Hours	12%
3.1	Construction of JFET, Operation of JFET, JFET Characteristics, Pinch-Off Voltage		
3.2	JFET Volt-Ampere Characteristics, FET Small Signal, MOSFET, FET as a VVR FET Biasing		
3.3	Fixed Bias Circuit, Voltage Divider Biasing Circuit, Self Bias Circuit, Biasing for Depletion type MOSFET		
3.4	JFET as an Amplifier, JFET Low Frequency Small Signal Model, Common Source Circuit, Common Drain Circuit		
4.	Multistage Amplifiers	7 Hours	12%
4.1	Classification of Amplifiers, Distortion in Amplifiers, Frequency Response of an Amplifier, Step Response of an Amplifier.		
4.2	Band - pass of Cascaded Stages, Two Stage RC Amplifier, High Frequency Response of Two Cascaded CE Transistor Stages, Multistage CE Amplifier Cascade At High Frequencies		
5.	Measurement Errors	3 Hours	6%
5.1	Introduction to Subject, Definitions		
5.2	Accuracy and Precision, Significant Figures, Types of Error		
5.3	Statistical Analysis, Probability of Errors, Limiting Errors		
6.	Bridge Measurements	5 Hours	10%
6.1	Introduction, Wheatstone Bridge, Kelvin Bridge		
6.2	AC Bridge And Their Applications, Maxwell Bridge, Hay's Bridge		
6.3	Unbalance Conditions, Wein Bridge, Schering Bridge		
7.	Digital Instruments	8 Hours	12%
7.1	Digital Frequency Meter, High Frequency Measurements		
7.2	Resolution & Sensitivity of Digital Meters		
7.3	LCR Meter and Q Meter		
7.4	Light Emitting Diode, Liquid Crystal Diodes		
8	Oscilloscope	4 Hours	8%
8.1	Introduction, Oscilloscope Block Diagram, Cathode Ray Tube		

8.2	Block Diagram Of DSO , Its Principle and Working, Advantages And Applications		
8.3	Special Oscilloscope		
9	Function Generator And Analyzer	4 Hours	8%
9.1	Introduction, The Sine Wave Generator		
9.2	Function Generator		
9.3	Logic Analyzer & Spectrum Analyzer		
10	Data Acquisition Systems and Transducers	6 Hours	10%
10.1	Objective of DAS, signal conditioning of the inputs, single channel DAS		
10.2	Electrical transducer, selecting a transducer, resistive transducer		
10.3	Strain gauges, thermistor, thermocouple		
10.4	LVDT, load cell,		
10.5	Piezoelectric transducer, photo electric transducer, photo voltaic cell		

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the need for basic Semiconductor physics for various applications and role of error in Measurement
CO2	Apply proper Transistor for required applications
CO3	Analyze Multistage Amplifiers and designing with real time examples.
CO4	Evaluate Bridge Measurement and implementation of the same.
CO5	Investigate about advancement in data acquisition system and applications of Transducer.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	3	3	-	-	-	-	-	-
CO2	3	2	3	3	3	1	-	-	-	-	-	-
CO3	3	3	2	3	1	3	-	-	-	-	-	-
CO4	3	3	3	3	3	3	-	-	-	-	-	-
CO5	-	-	-	-	-	3	-	1	2	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books & Text Books:

1. Electronics Devices and Circuit Theory (sixth Edition) by Boylestead & Louis Nashelsky published by Pearson Education Asia
2. Integrated Electronics by Jacob Millman Grable Tata McGraw-Hill Edition 1991
3. Modern electronics Instrumentation and measurement techniques by Albert D. Helfrick And William D. Cooper
4. Electronic Instruments and Measurements by David Bell
5. A course in Electrical and electronics measurement and instrumentation by A.K. Sawhney, 2nd Edition by Dhanpatrai.
6. Electronic instrumentation by H.S. Kalsi, 2nd Edition ,Tata McGraw Hill Publications

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC281.01: INTRODUCTION TO MATLAB PROGRAMMING
(UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	2
Marks	0	100	100	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to MATLAB Basics	3
2.	Basic MATLAB Functions	8
3.	Interactive computation	8
4.	Scripts and Functions in MATLAB	8
5.	Applications	3
	Total Hours (Theory):	0
	Total Hours (Lab):	30
	Total Hours:	30

Detailed Syllabus:

1.	Introduction to MATLAB Basics	3 Hours	10%
1.1	MATLAB windows, On-line help	1Hr	
1.2	Input- Output, File Types	1 Hr	
1.3	General commands to remember	1 Hr	
2.	Basic MATLAB Functions	8 Hours	25%
2.1	Working with Arrays of Number	1Hr	
2.2	Creating and Printing Simple Plots	1Hr	

2.3	Creating, Saving, and Executing a Script File	2Hrs
2.4	Creating and Executing a Function File	2Hrs
2.5	Working with Files and Directories	2Hrs
3.	Interactive computation	8 Hours 25%
3.1	Matrix and Vectors	1Hr
3.2	Matrix and Array Operations	3Hrs
3.3	Creating and Using Inline Functions	2Hrs
3.4	Using Built in Functions	2Hrs
4.	Scripts and Functions in MATLAB	8 Hours 25%
4.1	Scripts Files	2Hrs
4.2	Function Files	3Hrs
4.3	Language-Specific Features	3Hrs
5.	Applications	3 Hours 15%
5.1	Solving a Linear system	1Hr
5.2	Finding eigenvalues and eigenvectors	1 Hr
5.3	Matrix Factorizations	1 Hr

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Develop programming skill & simulation for engineering problems.
CO2	Importance of mathematics in MATLAB software implementation
CO3	Simplify basic mathematical, electrical, electronic problems in MATLAB
CO4	Able to simulate various discipline applications using MATLAB

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	-	-	1	-	-	-	-	-	-	1
CO2	3	3	1	1	1	-	-	-	-	-	-	1
CO3	3	2	-	1	1	-	-	-	-	-	-	1
CO4	3	3	1	1	1	-	-	-	-	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Getting Started with MATLAB, Rudrapratap (IISc, Bangalore), Oxford University Press.

2. A Guide to MATLAB, Brian R Hunt, Ronald L Lipsman, Cambridge University Press.

❖ Web Materials/ Reading Material:

1. Lecture Notes
2. Hand Outs
3. Chapter wise Assignments

Simulators:

MATLAB

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS121.01A - Creativity, Problem Solving and Innovation

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact	Theory		Practical		Total
				Hours/Week	Internal	External	Internal	External	
III	HS121.01A	Creativity, Problem Solving and Innovation	02	02	--	--	30	70	100

Course Outline

Module No	Title/Topic	Contact Hours
1	Introduction to Creativity, Problem Solving and Innovation - Definitions of Creativity and Innovation - Need for Problem Solving and Innovation - Scope of Creativity in various Domains - Types and Styles of Thinking - Strategies to develop Creativity, Problem Solving and Innovation skills	06
2	Questioning, Learning and Visualization - Strategy and Methods of Questioning - Asking the Right Questions - Strategy of Learning and its Importance - Sources and Methods of Learning - Purpose and Value of Creativity Education in real life - Visualization strategies - Making thoughts Visible - Mind Mapping and Visualizing Thinking	06
3	Creative Thinking and Problem Solving ☐ Creative Thinking and its need ☐ Strategy of Thinking Fluency ☐ Generating all Possibilities ☐ SCAMPER Technique ☐ Divergent Vs Convergent Thinking	06

- ☐ Lateral Vs Vertical Thinking
 - ☐ Fusion of Ideas for Problem Solving
 - ☐ Applying strategies for Problem Solving
- 4 **Logic, Language and Reasoning** 06
- Basic Concepts of Logic
 - Statement Vs Sentence
 - Premises Vs Conclusion
 - Concept of an Argument
 - Functions of Language: Informative, Expressive and Directive
 - Inductive Vs Deductive Reasoning
 - Critical Thinking & Creativity
 - Moral Reasoning
- 5 **Contemporary Issues and Practices in Creativity and Problem Solving** 06
- Cognitive Research Trust Thinking for Creatively Solving Problems
 - Case Study on Contemporary Issues and Practices in Creativity and Problem Solving

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Puzzles, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc

IV. Evaluation:

There will be no end semester university examinations. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation. The concerned teacher shall evaluate students distribute the marks (out of 30 as Internal and out of 70 as External) and submit them.

Evaluation Scheme

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Attendance	100 %	--	20
2	Individual Activity Participation	As stipulated by the Resource Person(s) in the Training		20
3	Group Activity Participation			20
4	Presentation			30
5	Feedback on Improvement			10
Total				100

Course Outcome (COs):

After completion of the course, the student would:

CO1	Demonstrate creativity in their day to day activities and academic output
-----	---

CO2	Solve personal, social and professional problems with a positive and an objective mindset
CO3	Think creatively and work towards problem solving in a strategic way
CO4	Initiate new and innovative practices in their chosen field of profession

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	1	-	-	-	-	-	-	-	-	-
CO2	1	2	3	1	-	-	-	-	-	-	-	-	-	2
CO3	-	-	-	-	1	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	1	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Books / Reference Books / Reading

Text Books

1. R Keith Sawyer, Zig Zag, The Surprising Path to Greater Creativity, Jossy-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

Reference Books

3. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
4. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
5. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
6. Tony Buzan, How to Mind Map, Thorsons Publication 2002
7. Scott Berkum, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
8. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
9. Ira Flatow, The all Laughed, Harper Publication 1992
10. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

Additional Readings

11. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007

12. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
13. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
14. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
15. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
16. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
17. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
18. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
19. Soctt G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
20. Donald J Treffinger, scott G Isaksen, K Brain stead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
21. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
22. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
23. Howard Gardner, Creating minds, Basic Books Publication 1993
24. Mihaly Csikzentmihalyi, Creativity–Flow and Psychology of Discovery and Invention, Harper Publication 1996
25. Martin Gerdner, W. H., Ahal Insight, Freeman Publication 1978
26. Paul Sloane, Test Your Lateral Thinking IQ, Sterling Publication 1994
27. Paul Sloane & Des Machale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

Articles / Videos / Other Suggested Materials

- Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

**B. Tech. (Electronics and Communication Engineering)
Programme**

SYLLABI
(Semester – 4)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC263: CONTROL THEORY

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction	5
2.	Mathematical Models of Physical Systems	5
3.	Stability	6
4.	Time Domain Analysis	6
5.	Frequency Response	7
6.	Root Locus Method	6
7.	Nyquist stability criterion	7
8.	PID Controllers and its applications	3
	Total Hours (Theory):	45
	Total Hours (Lab):	0
	Total Hours:	45

Detailed Syllabus:

1. Introduction	5 Hours (7 Marks)	10%
1.1 Introduction & Classification of Control Systems, Transfer function, Block diagram reduction, Signal flow graphs		
2. Mathematical Models of Physical Systems	5 Hours (7 Marks)	10%
2.1 Differential Equations of Physical Systems, Transfer Functions of Mechanical Systems & Electrical Systems		
2.2 Analogous System: Analogous In Force(Torque)-Voltage Analogy & Force(Torque)-Current Analogy		
3. Stability	6 Hours (11 Marks)	16%
3.1 Concept of Stability, necessary and sufficient conditions of stability		
3.2 Hurwitz Stability Criterion		
3.3 Routh Stability Criterion		
4. Time Domain Analysis	6 Hours (11 Marks)	16%
4.1 Typical test Signals		
4.2 Time Response Specifications		
4.3 Steady State Error and Error Constants		
5. Frequency Response	7 Hours (7 Marks)	9%
5.1 Advantages and limitations of frequency response analysis		
5.2 Bode plots		
5.3 Correlation between time response and frequency response		
6. Root Locus Method	6 Hours (9 Marks)	13%
6.1 The Concept of Root Locus		
6.2 Rules For Constructions of Root Locus		
7. Nyquist stability criterion	7 Hours (14 Marks)	20%
7.1 Development of the criterion		
7.2 Gain and phase margins		
8. PID Controllers and its applications	3 Hours (4 Marks)	6%
8.1 Proportional, derivative and Integral controllers		
8.2 Applications of PID Controllers		

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the different types of control systems in real time applications and its classifications.
CO2	Analyse the transfer functions of mechanical and electrical systems, and apply its reduction techniques.
CO3	Apply different stability criterions to acquire the stability information of open loop and closed loop control system.
CO4	Analyse the time domain response of test inputs.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	--	-	-	-	-	-	-
CO2	3	3	3	3	3	-	-	-	3	-	-	-
CO3	2	3	3	2	3	-	-	-	3	-	-	-
CO4	2	3	3	3	3	-	-	-	3	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Control System Engineering. by I.J. Nagrath & M. Gopal, New Age International Publishers, New Delhi.
2. Control System Engineering by U.A Patel, Mahajan Publication
3. Principles of Control Systems by U.A Patel & S.C Goyal, Technical Publication
4. Linear Control Systems By B.S.Manke, khanna Publication
5. Modern Control Engineering by Ogata K. 4th Edn. Prentice Hall.

❖ Web Materials/ Reading Material:

1. NPTEL lecture series
2. Lab Manual
3. Assignments
4. Question Bank
5. Hand Outs
6. NPTEL lecture series

❖ Simulators:

1. MATLAB

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC256: ANALOG CIRCUITS & APPLICATIONS

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

Out Line of the Course:

Sr. No.	Title of The Unit	Minimum Number of Hours
1.	Power Supply	02
2.	Power Amplifier	07
3.	Feedback Amplifier	06
4.	Oscillators	04
5.	Non-Linear Wave Shaping Circuits	04
6.	Operational Amplifier Characteristics	09
7.	Linear and Non Linear Applications of Op-Amp	09
8.	Active Filters	05
9.	Specialized ICs	03
10.	Data Converters	06
11.	Contemporary topics in Electronics	05

Total Hours (Theory): 60

Total Hours (Lab): 30

Total: 90

Detailed Syllabus:

1. Power Supply	02 Hours	5%
1.1 Regulated Power Supply – Series Voltage Regulator Design,		
1.2 Short Circuit and Overload Protections, Voltage Regulator ICs.		
2. Power Amplifier	06 Hours	10%
2.1 AC/DC Load Line, Class A, Second Harmonics Distortion, Higher Order Harmonics		
2.2 Transformer-Coupled Audio Power Amplifiers, Push-Pull Amplifier, Efficiency		
3. Feedback Amplifiers	06 Hours	10%
3.1 Feedback Concept, Transfer Gain With Feedback, General Characteristics of Negative-Feedback Amplifiers, Input Resistance, Output Resistances		
3.2 Current-Shunt Feedback, Voltage-Shunt Feedback, Current-Series Feedback, Voltage-Series Feedback.		
4. Oscillators	04 Hours	10%
4.1 Sinusoidal Oscillator, Phase Shift Oscillators, Resonant-Circuit, Hartley Oscillators, Colpitts Oscillators.		
4.2 Wien Bridge Oscillators, Crystals Oscillator		
5 Non-Linear Wave Shaping Circuits	04 Hours	10%
5.1 Clipping Circuits: Diode Clippers - Series and Shunt Clipper, Clipping at Two Independent Levels, Transistor Clipper		
5.2 Clamping Circuits: Positive and Negative Clamper, Biased Clampers		
6. Operational Amplifier and Characteristics	09 Hours	12%
6.1 Operational Amplifier Symbols and terminals, DC power supply for Op-Amp, Ideal Differential Amplifier, Differential gain, Common Mode Gain, Common Mode Rejection Ratio.		
6.2 Block diagram of Typical Op-Amp, Equivalent Circuit of Op-Amp, Ideal Op-amp, Important Characteristics of Ideal Op-Amp, Characteristics of Practical Op-Amp, and OP-AMP IC 741.		

6.3	Open Loop Configuration of Op-Amp, Voltage Transfer Characteristics of Op-Amp, Open Loop Configuration of Ideal Op-Amp, Differential Amplifier, Inverting Amplifier and Non-Inverting Amplifier in Open Loop mode.		
6.4	Op-Amp in Closed Loop Configuration, Concept of Virtual Short, Virtual Ground, Zero Input Current, Closed Loop and Open Loop Op-Amp Applications, Inverting Amplifier, Non-Inverting Amplifier, Voltage Follower.		
7.	Linear and Non Linear Applications of Operational Amplifier	8 Hours	12%
7.1	Summing, Scaling and Averaging Amplifier using Inverting and Non-Inverting Amplifier.		
7.2	Difference Amplifier, Subtractor, Integrator, Differentiator.		
7.3	Zero Crossing Detector, Clipper and Clampers		
7.4	Peak Detector, The Op-Amp as Voltage Comparator, Schmitt Trigger Circuit		
8	Filters Design using Operational Amplifier	05 Hours	8%
	Classification of filters, Magnitude and frequency Scaling,		
8.1	magnitude and attenuation characteristics of ideal and practical filters		
8.2	Passive filter realization, Cascade design, Design Parameter Q & ω_0 , Biquad (Universal) filter design		
9	Specialized IC	03 Hours	8%
	555 Timer and its Applications: Introduction, Basics of Timer IC		
9.1	555, Block Diagram of IC 555, Astable and Monostable Multivibrator, Applications as Frequency Divider, Square Wave Generator, Free-Running Ramp Generator etc		
10	Data Converters	06 Hours	10%
10.1	Digital To Analog Conversion, R-2R Ladder Type DAC, Weighted Resistor Type DAC		
10.2	Analog To Digital Conversion, Counter Type A/D Converter, Tracking Type A/D Converter, Flash-Type A/D Converter, Successive Approximation Type ADC.		

11 Contemporary topics in Electronics**05 Hrs****5%**

11.1 Introduction to Memristors

11.2 Introduction to Flexible Electronics

11.3 Introduction to Nanoelectronics

Course Outcomes (CO):**At the end of the course student will be able to:**

CO1	Explain working principle of various electronic circuits and their application in real life.
CO2	Understand and apply of analog to digital converter, digital to analog converter in various applications.
CO3	Analyze different signal processing circuit and the use in industrial, real life, modern control system application.
CO4	Select appropriate semiconductor devices depending upon application considering commercial and technology up-gradation.
CO5	Design various circuits with the help of operational amplifier, timer IC and Check the properties of amplifiers and decide to apply them in proper field.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	3	3	-	-	2	2	-	-	2
CO3	3	3	3	3	3	-	-	2	2	-	-	2
CO4	3	3	3	2	-	-	-	2	-	-	-	2
CO5	3	3	3	3	3	-	-	2	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:**❖ Reference Books:**

1. Electronics Devices and Circuits Theory (Sixth Edition) By Boylestead & Nashelsky
PHI publication
2. Ramakant A. Gayakwad ,Op-Amp and Linear Integrated Circuits , 4th Edition, PHI
Publications
3. Van Valkenburg ,Analog filter design , Oxford Publication
4. J. Michael Jacob ,Application and design with Analog ICs 2nd Edition, PHI Publication
5. Sergio Franco ,Design with Operational Amplifiers &Analog ICs 3rd Edition , TMH
Publication

❖ **Reading Materials, Web Materials With Full Citations**

1. Lecture Notes
2. Hand Outs
3. Assignments
4. Project and Quiz

❖ **Simulators**

1. LTspice
2. Proteus Design Suite

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC257: MICROPROCESSOR & INTERFACING

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	8086 Processor Architecture	05
2	8086 Assembly Language Programming	10
3	Interrupt Programming	05
4	Basic Peripherals & their Interfacing with 8086	10
5	Instruction Set Architecture	05
6	The Arithmetic Logic Unit	05
7	Data Path & Control	05
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

- | | | | |
|----------|--------------------------------------|-----------------|------------|
| 1 | 8086 Processor Architecture | 05 Hours | 11% |
| 1.1 | Architecture of 8086 Architecture | | |
| 1.2 | Signal Description of 8086 Processor | | |
| 1.3 | Physical Memory Organization | | |
| 1.4 | General Bus Operation | | |

1.5	Minimum & Maximum mode of 8086 Microprocessor		
2	8086 Assembly Language Programming	10 Hours	22%
2.1	Machine Language Instruction Format		
2.2	Addressing mode of 8086		
2.3	Instruction Set of 8086		
2.4	Assembler Directives and Operators		
2.5	Assembly Language Example Program		
3	Interrupt Programming	05 Hours	11%
3.1	Stack Structure of 8086		
3.2	Interrupt & Interrupt Service Routine		
3.3	Interrupt Cycle, Non Mask able & Mask able Interrupt		
3.4	Interrupt Programming		
3.5	Passing Parameters to Procedures & Macros		
4	Basic Peripherals & their Interfacing with 8086	10 Hours	23%
4.1	Interfacing & Programming of 8255		
4.2	Interfacing & Programming of ADC		
4.3	Interfacing & Programming of 8254 & 8259		
4.4	Interfacing & Programming of 8279 & 8251		
4.5	Interfacing & Programming of 8257		
5	Instruction Set Architecture	05 Hours	11%
5.1	Instructions & Addressing		
5.2	Procedures & Data		
5.3	Execution flow of Assembly Language Programs		
5.4	Instruction Set Variations		
6	The Arithmetic Logic Unit	5 Hours	11%
6.1	Number Representation		
6.2	Adders & Simple ALUs		
6.3	Multipliers & Dividers		
6.4	Floating Point Arithmetic		
7	Data Path & Control	05 Hours	11%
7.1	Instruction Execution Steps		
7.2	Control Unit Synthesis		

7.3 Pipelined Data Paths

7.4 Pipeline Performance Limits

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the Architecture of 8086 Microprocessor
CO2	Able to Understand Special features of 8086 Microprocessor
CO3	Perform real time applications on 8086 Microprocessor
CO4	Identify the elements of modern instructions sets and their impact on processor design
CO5	Able to learn the concept of Pipelining in Microprocessor

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	1	-	-	-	-	-	1	-	1
CO2	2	3	3	2	3	-	-	-	2	-	-	-
CO3	3	3	3	2	-	-	-	-	-	-	-	-
CO4	2	3	3	3	3	1	1	-	2	-	-	-
CO5	3	3	2	3	2	1	1	-	2	-	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Advanced Microprocessors & Peripherals, 3rd Edition, A.K. Ray & K.M. Bhurhandi, McGraw-Hill
2. Computer Architecture by Behrooz Parhami, OUP USA

❖ Web Materials/ Reading Material:

1. Microprocessors & Interfacing, Programming & Hardware by Douglas V. Hall, Tata McGraw-Hill Publication
2. Computer System Architecture by Morris Mano, Tata McGraw-Hill Application
3. Hand Outs
4. Assignments
5. Question Bank

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC258: SOFTWARE MODELLING - I

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	4	4	2
Marks	0	100	100	

A. Objective of the Course:

The Objective of the Course is to teach students the fundamentals of Linux Operating system, Linux command with sophisticated editors and shell scripting. Also aware students regarding TCL scripting language with basic syntax and datatypes.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Linux Fundamental	08
2.	Linux Editors: Vi & Vim	10
3.	Shell Scripting	12
4.	Basics of TCL (Tool Command Language)	08
5.	TCL Operators, Decisions, Loops	10
6.	TCL Array, Strings, Lists, Dictionary	12
	Total Hours (Theory):	00
	Total Hours (Lab):	60
	Total Hours:	60

C. Detailed Syllabus:

1.	Linux Fundamentals	08 Hours	10 %
1.1	Introduction to Linux		
1.2	Linux Distribution		
1.3	Linux Commands		
1.4	Linux File Structure		
2.	Linux Editors: Vi & Vim	10 Hours	20 %
2.1	Outline of Various Linux Editors		
2.2	Introduction to Vi & Vim Editors		
2.3	Difference Between Vi & Vim Editors		
2.4	Commands for Vi modes		
2.5	Command for Vim modes		
3.	Shell Scripting	12 Hours	20%
3.1	Introduction to Shell		
3.2	Shell Types and Variables		
3.3	Loops and Control Statement		
3.4	Command Line Arguments		
4	Basics of TCL (Tool Command Language)	08 Hours	10 %
4.1	Overview of TCL		
4.2	Environment Setup in TCL		
4.3	Variables and Special Variables in TCL		
4.4	Datatypes in TCL		
5	TCL Operators, Decisions, Loops	10 Hours	20 %
5.1	TCL Operators		
5.2	TCL Decision statements		
5.3	TCL Loop Syntax		
6	TCL Array, Strings, Lists, Dictionary	12 Hours	20 %
6.1	TCL Array Methods		

6.2	TCL Strings Methods		
6.3	TCL Lists Methods		
6.4	TCL Dictionary Methods		

Course Outcomes (CO):

At the end of the course student will be able

CO1	To Evaluate Linux as an Operating System
CO2	To Use Linux command line to complete a series of tasks
CO3	To demonstrate basic shell scripting
CO4	To understand the fundamentals of TCL
CO5	To design script based on TCL

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	-	3	-	-	-	-	-	1	1
CO2	3	2	3	-	3	-	-	-	-	-	1	1
CO3	3	-	3	-	3	-	-	-	-	-	-	-
CO4	3	2	3	-	3	-	-	-	-	-	1	1
CO5	3	2	3	-	3	-	-	-	-	-	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Fundamentals of Linux by Olvir Pelz, Packt Publishing

❖ Web Materials/ Reading Material:

1. Application Oriented TCL for VLSI
2. Hand Outs
3. Assignments
4. Question Bank

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC259: MINI PROJECT-II

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	1
Marks	-	50	50	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Arduino based project implementation	15
2.	Hardware/ Module Development	15
	Total Hours (Theory):	00
	Total Hours (Lab):	30
	Total Hours:	30

Detailed Syllabus:

The student can carry the project work individually. The student will be given choice to select the project from areas of Electronics & Communication engineering after discussions with instructor. The topic should be related to the field of electronics and Communication engineering. It is recommended to carry out innovative projects.

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand project related algorithm/circuit/configuration.
CO2	Apply implementation method and list components or parts required.
CO3	Analyze practical difficulties faced or observed during implementations of

	project and discover solutions for them.
CO4	Analyze results of projects with other similar design specifications.
CO5	Create technical report which defines the designed project in details.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	-	-	3	-	-	-
CO2	3	3	3	3	3	3	-	-	3	-	3	-
CO3	3	2	2	3	3	2	-	-	3	-	2	-
CO4	3	-	-	-	-	-	-	-	3	-	2	3
CO5	-	-	-	-	-	-	-	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material

❖ Reference Books:

1. Exploring Arduino: Tools and Techniques for Engineering Wizardry 1st Edition by Jeremy Blum, Wiley Press, 1st Edition
2. Adventures in Arduino by Becky Stewart, Wiley Press, 1st Edition
3. Programming Arduino: Getting Started with Sketches, McGraw Hill by Simon Monk, Second Edition

❖ Web material:

- <https://www.arduino.cc/en/pmwiki.php?n=Guide/References>
- <https://create.arduino.cc/projecthub/projects/tags/arduino>
- <https://www.electronicshub.org/arduino-project-ideas/>

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC271: HARDWARE MODELING USING VERILOG

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	50	50	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to digital circuit design flow	04
2.	Verilog variables, operators and language constructs	06
3.	Modeling combinational circuits using Verilog	08
4.	Modeling sequential circuits using Verilog	08
5.	Verilog test benches and design simulation	04
	Total Hours (Lab):	30
	Total Hours:	30

Detailed Syllabus:

- | | | | |
|-----------|---|----------------|------------|
| 1. | Introduction to digital circuit design flow | 4 Hours | 13% |
| 1.1 | FPGA & ASIC VLSI Design Flow, Semi-custom Design Styles, Full Custom Design Styles | | |
| 1.2 | Stick Diagram, Steps in Physical Design Flow | | |
| 2. | Verilog variables, operators and language constructs | 6 Hours | 20% |
| 2.1 | Verilog Language Standard & features, Data types, Control Statements | | |
| 2.2 | Verilog Operators & Verilog Modelling Styles : Dataflow, Structural, Behavioral, Gate level | | |
| 2.3 | Verilog variables, operators and language constructs | | |
| 3. | Modeling combinational circuits using Verilog | 8 Hours | 27% |

- 3.1 Procedural Statements, Blocking & Non- Blocking Statements, User Defined Primitives
- 3.2 Model combinational circuits : Adder, Subtractor, Decoder, Encoder, Multiplexer, ALU
- 4. Modeling sequential circuits using Verilog** **8 Hours 27%**
- 4.1 Latches, Flip-Flops, Counters, Shift Registers, Memory
- 4.2 Finite State Machines (Moore & Mealy), Vending Machines, Sequence detectors
- 5. Verilog test benches and design simulation** **4 Hours 13%**
- 5.1 Test bench Architecture, Simulation Directives,
- 5.2 Test bench for all combinational and Sequential circuits

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the need and trends in VLSI Design & Hardware Description Language.
CO2	Demonstrate and apply various Verilog modelling styles and various Language constructs.
CO3	Model combinational logic circuits using HDL and verify its functionality
CO4	Analyse the architecture of Verilog Testbench and Design testbenches for various digital logic circuits.
CO5	Design sequential logic circuits using HDL and verify its functionality

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	-	1	1	1	2
CO2	3	-	1	-	3	-	-	-	1	1	1	2
CO3	3	3	3	2	3	-	-	-	1	1	1	2
CO4	3	3	3	2	3	-	-	-	1	1	1	2
CO5	3	3	3	2	3	-	-	-	1	1	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Verilog HDL: A Guide to Digital Design and Synthesis, Second Edition By Samir

Palnitkar

2. Design Through Verilog HDL by T R Padmanabhan, WILEY INDIA
3. Digital System Design Using VHDL, Charles H. Roth, Thomson, 2002

❖ Web Materials/ Reading Material:

6. [http:// https://accelera.org/images/downloads/standards/v-ams/VAMS-LRM-2-4.pdf](http://https://accelera.org/images/downloads/standards/v-ams/VAMS-LRM-2-4.pdf)
7. Lab Manuals
8. Hand Outs
9. Assignments
10. Question Bank

Simulators:

Xilinx Altera

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC272: INTRODUCTION TO ROBOTICS

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	50	50	

Outline of the Course:

The course will start with a brief introduction to robots and robotics. The motivation behind keeping robots in modern industries will be discussed. After providing a brief history of robotics, different components of a robotic system will be identified. The method of determining degrees of freedom of a robotic system will be discussed with some examples. After classifying the robots based on certain criteria, workspace analysis of manipulators will be carried out. Applications of robots in different areas like in manufacturing units, medical science, space, and others, will be discussed. Various methods of robot teaching will be explained with some suitable examples.

Total Hours (Lab) : 30

Total Hours (Theory): 0

Total Hours:30

C. Detailed Syllabus:

1 Introduction to Robots and Robotics	05 Hrs
1.1 Robots, Robotics, Application of Robotics, Types of Robot	01
1.2 Componets of Robotic System, Degrees of Freedom of Robotic System	02
1.3 Specifications of Robot	
2 Robot Kinematics	06 Hrs
2.1 Representation of Position &Orientation, Translation Operator, Rotational Operator	04

2.2 Forward Kinematics, Inverse Kinematics	02
3 Trajectory Planning	03 Hrs
3.1 Linear Trajectory Function, Singularity Check using Jacobian	
4 Robot Dynamics	06 Hrs
4.1 Determination of Joint torque/force for various types of link : Inertia Term, Centrifugal term, Gravity Term	03
4.2 Determination of Potential & Kinetic Energy of Manipulator	02
4.3 Control Architecture	01
5 Sensors	04 Hrs
5.1 Introduction, Characteristics & Classification of Sensors used in Robots	
5.2 Position Sensor, Force (Moment Sensor), Range Sensor, Proximity Sensor	
6 Robotic Vision	02 Hrs
6.1 Capturing Image, Image preprocessing, Boundary descriptors	
7 Robot Motion Planning	04 hrs
7.1 Free space motion planning, Compliant motion planning, Motion Planning Schemes	

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the principles of robotics.
CO2	Analyze the robot kinematic position, dynamic actuator force and Create Mechatronic modeling and design
CO3	Evaluate and validate the simulation models
CO4	Apply the concept of sensors used in robotics application & basic robot programming techniques

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	2	-	-	-	-	-	-	2
CO2	3	3	3	3	3	-	-	-	3	-	-	-
CO3	3	3	3	3	3	-	-	-	2	-	-	-
CO4	3	3	3	3	3	2	-	-	2	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Instructional Method and Pedagogy:

The Course instructor will follow instruction methodology and pedagogy as follows:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Attendance is compulsory in laboratory which carries a 10% component of the overall evaluation.
- In each subsequent lab student have to regularly complete the assigned task.
- Student/Group has to submit a lab report. Evaluation will be made according to performance in subsequent lab sessions.

Student Learning Outcomes:

After successfully completing this course, the student shall be able to:

- Understand need and application of Robotics.
- Design Kinematics, trajectory planning and dynamics of robots.
- Case study of Wheeled robot
- Interfacing sensors to robots.

F. Reference Books:

1. Fundamentals of Robotics by D.K. Pratihar, Narosa Publishing House, New-Delhi, 2017
2. Robotics by K.S. Fu, R.C. Gonzalez, C.S.G. Lee, McGraw-Hill Book Company, 1987
3. Introduction to Robotics by J.J. Craig, Addison-Wesley Publishing Company, 1986

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC273: FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	50	50	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Artificial intelligence: History, Trends and Future	04
2.	Problem Solving by search	10
3.	Knowledge Representation and Reasoning	06
4.	Reasoning under uncertainty	05
5.	Planning	05
	Total Hours (Theory):	0
	Total Hours (Lab):	30
	Total Hours:	30

Detailed Syllabus:

1.	Artificial intelligence: History, Trends and Future	4 Hours	13%
1.1	Introduction to Artificial Intelligence (AI), Evolution of AI		1Hr
1.2	AI technique, Production System : Components & Control Strategies		2 Hr
1.3	Problem Solving as State Space Search : Uninformed Search		1 Hr
2.	Problem Solving by search	10 Hours	33%
2.1	Heuristic Search: Heuristic Function, Problem Relaxation		3 Hr
2.2	Informed Search, Best first search, Graph Search		3 Hr

2.3	Constraint Satisfaction Problems : N Queens Problem , Cryptarithmic Puzzle	2 Hrs
2.4	Searching AND/OR graphs, Game playing	2 Hrs
3.	Knowledge Representation and Reasoning	6 Hours 20%
3.1	Introduction, Propositional logic, First order Logic	2 Hr
3.2	Inference in First order Logic, Answer Extraction	2 Hr
3.3	Procedural control of reasoning	2 Hrs
4.	Reasoning under uncertainty	5 Hours 17%
4.1	Introduction, Bayesian Network, Decision network	5 Hrs
5.	Planning	5 Hours 17%
5.1	Introduction, Planning space graph, Planning & Acting	5 Hrs

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Define the fundamental of Artificial Intelligence (AI)
CO2	Understand the basic concept of problem solving approach
CO3	Develop a basic understanding of problem solving, knowledge representation, reasoning and learning methods of AI

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	-	-	1	-	-	-	-	-	-	1
CO2	3	3	2	1	1	-	-	-	-	-	-	1
CO3	3	3	3	2	1	-	-	-	-	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Patrick Henry Winston, Artificial Intelligence, Third Edition, Addison Wesley Publishing Company, 2004
2. Nils J Nilsson, Principles of Artificial Intelligence, Illustrated Reprint Edition, Springer Heidelberg, 2014.
3. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 3rd

Edition, PHI 2009.

4. Nils J. Nilsson, Quest for Artificial Intelligence, First Edition, Cambridge University Press, 2010.

❖ Web Materials/ Reading Material:

1. Lecture Notes NPTEL
2. Chapter wise Quiz

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC275: PROJECT MANAGEMENT

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	50	50	

. Outline of the Course:

With the concept of managing Big Projects under costs and time constraints, it is imperative, that people working in manufacturing/process/service industry have a very good understanding of the general and advanced concepts of Project Management. It is with this motivation that this course is designed, to meet the demand in the market from, UG to PG students coming from a variety of fields, be it Engineering or Management.

Total Hours (Lab) : 30

Total Hours (Theory): 0

Total Hours:30

Detailed Syllabus:

1	Project Management, Concepts and Definitions;	04 Hrs	14%
1.1	Introduction, Project Management Process,		03 Hrs
1.2	Project Management Cycle		01 Hrs
2	Risk Management	06 Hrs	20%
2.1	Project Stakeholder , Project Risk management		02 Hrs
2.2	Risk associated with Projects Decision, Decision Tree Modeling		02 Hrs
2.3	Project Risk Management Analysis, CAPM Model		02 Hrs
3	Utility Theory	06 Hrs	20%
3.1	Introduction		01 Hrs

3.2	Application of Utility Theory in Project Management	04 Hrs
3.3	Work Breakdown Structure, Cost Estimation in Project Management	01 Hrs
4	Program Evaluation Review Technique (PERT)	06 Hrs 20%
4.1	Introduction	01 Hrs
4.2	Distribution, Calculation, Forward Pass Method	02 Hrs
4.3	GANNT Chart and Precedence Diagrams CPM	02 Hrs
4.4	CPM Method	01 Hrs
5	Project Scheduling and Control	08 Hrs 26%
5.1	Concepts of Scheduling, Forward & Backward Scheduling	03 Hrs
5.2	Graphical Evaluation and Review Technique (GERT)	03 Hrs
5.3	Q-GERT	02 Hrs

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the basics of Project Management process and cycle.
CO2	Understand the principles of Project risk management and evaluate the risk of the real time projects using the models.
CO3	Apply the principles of Utility theory, PERT and CPM.
CO4	Analyse the learning and understand the techniques for Project planning, scheduling and Execution Control.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1	0	0	0	1	0	0	2	2	3	2
CO2	1	2	1	1	2	2	1	0	2	2	3	3
CO3	1	2	2	2	1	2	1	1	3	3	3	2
CO4	1	3	3	2	1	2	1	1	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Instructional Method and Pedagogy:

The Course instructor will follow instruction methodology and pedagogy as follows:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Attendance is compulsory in laboratory which carries a 10% component of the overall evaluation.
- In each subsequent lab student have to regularly complete the assigned task.
- Student/Group has to submit a lab report. Evaluation will be made according to performance in subsequent lab sessions.

E. Student Learning Outcomes:

After successfully completing this course, the student shall be able to:

- Understand concept of Project Management and Project Management cycle
- Comprehend Decision making using tree modeling
- Understand Cost Evaluation Techniques in Project Management
- Learn importance of PER and CPM project life cycles

F. Reference Books:

1. Chandra, P., Projects, Tata McGraw-Hill Education, 2009.
2. Levy, F. K. and Wiest, J. D., A Management Guide to PERT/CPM, Prentice Hall, 1969.
3. Lewis, R., Project Management, McGraw-Hill, 2006.
4. Moder, J. J. and Phillips, C. R., Project Management With CPM, PERT and Precedence Diagramming, Van Nostrand Reinhold, 1983.
5. Morris, P. W. G., and Pinto, J. K., The Wiley Guide to Managing Projects, 2004, JohnWiley & Sons.

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND
COMMUNICATION
EC282.01: PROTOTYPING ELECTRONICS WITH ARDUINO
(UNIVERSITY ELECTIVE)

—

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	2
Marks	0	100	100	

A. Objective of the Course:

The objective of this course is to teach students regarding Arduino architecture and aware students regarding interfacing sensors and actuators with Arduino and use of Arduino C language.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Getting Started with Arduino	02
2.	Arduino C	06
3.	GPIO Programming	08
4.	Sensors & Actuators	06
5.	Wireless Devices	08
	Total Hours (Theory):	30
	Total Hours (Lab):	00
	Total Hours:	30

C. Detailed Syllabus:

- | | | |
|--|-----------------|------------|
| 1. Getting Started with Arduino | 02 Hours | 10% |
| 1.1 Introduction to Arduino | | 1 Hr |

1.2	Arduino Variants		
1.3	The Arduino UNO Board	1 Hr	
1.4	Electronics Components		
2.	Arduino C	06 Hours	20%
2.1	C Data types	2 Hrs	
2.2	Control Statements		
2.3	Functions & Arrays		
2.4	Using Modifying & Creating Arduino Libraries	2 Hrs	
2.5	Advanced Coding & Memory Handling in Arduino	2 Hrs	
3.	GPIO Programming	08 Hours	25%
3.1	LED Interfacing with Arduino	1 Hr	
3.2	Driving 7 Segment Display using Arduino	1 Hr	
3.3	Serial Communications & Soft serial Utility	2 Hrs	
3.4	LCD Interfacing using Arduino	1 Hr	
3.5	Matrix Keypad Interfacing using Arduino	1 Hr	
3.6	Controlling Speed of DC Motor using PWM Techniques	1 Hr	
3.7	Driving Stepper Motor using Arduino	1 Hr	
4.	Sensors & Actuators	06 Hours	20%
4.1	Introduction to Sensors & Actuators	2 Hrs	
4.2	Digital On/Off Sensors		
4.3	Analog Sensors	2 Hrs	
4.4	Pulse Sensors	2 Hrs	
5.	Wireless Devices	08 Hours	25%
5.1	Introduction to Wireless Devices	2 Hrs	
5.2	Interfacing GSM & GPS Module with Arduino		
5.3	Interfacing Bluetooth Module with Arduino	2 Hrs	
5.4	Interfacing Zigbee Module with Arduino	2 Hrs	
5.5	Interfacing IR Module with Arduino	2 Hrs	

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand about open source platform.
CO2	Understand the functionality of various input, output devices and analog and digital sensors.
CO3	Understand inbuilt serial communication facilities of arduino like I2C,UART and SPI.
CO4	Develop ArduinoC Programing skills.
CO5	Develop library files for sensors and actuators

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	3	-	-	1	-	-	-	-
CO2	3	3	-	2	3	-	-	2	2	-	-	3
CO3	1	1	3	-	3	-	-	-	2	1	-	-
CO4	-	-	-	-	3	-	-	-	2	-	-	3
CO5	3	-	-	2	3	-	-	-	3	2	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ **Reference Books:**

1. Arduino Cookbook, Michael Margolis, O'Reilly

❖ **Web Materials/ Reading Material:**

1. Beginning of Android ADK with Arduino by Mario
2. Hand Outs
3. Assignments
4. Question Bank

❖ **Simulators**

1. Arduino IDE
2. Proteus Design Suite

HS111.02 A: HUMAN VALUES & PROFESSIONAL ETHICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/01	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

- Ethical Leadership through Giving Voice to Values
<https://www.coursera.org/learn/uva-darden-giving-voice-to-values?skipBrowseRedirect=true>

Objectives of the Course:

To facilitate learners to:

- Develop a familiarity with the fundamental human values and professional ethics
- Understand basic concepts of values and ethics
- Explore and understand values, ethics in context of professional, social and personal spectrum
- Explore and understand values, ethics in context of globalization and global issues
- Explore an application of values and ethics in personal, social, academic, global and profession life.
- Facilitate the learners to understand harmony at all the levels of human living, and live accordingly.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Elements and Principles of Values	08
3.	Applied Ethics	08
4.	Value, Ethics & Global Issues	05
5.	Contemporary Issues in Values and Ethics	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

- | | | |
|--|---------|-----|
| 1. Introduction to Values and Ethics | 5 Hours | 17% |
| Need, Relevance and Significance of Values General, Concept and Meaning of Values and Ethics | | |
| 2. Elements and Principles of Values | 8 Hours | 26% |
| Universal & Personal Values, Social, Civic & Democratic Value | | |
| 3. Applied Ethics | 8 Hours | 26% |
| Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics | | |
| 4. Value, Ethics & Global Issues | 5 Hours | 17% |
| Cross-Cultural Issues, Role of Ethics & Values in Sustainability | | |
| 5. Contemporary Issues in Values and Ethics | 4 Hours | 14% |
| Case Studies, Presentations, Projects | | |

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the concepts and mechanics of values and ethics.
CO2	Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
CO3	Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
CO4	Develop their responsibility towards society.
CO5	Comprehend their own core values and adhere to those values at their workplace.
CO6	Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	2	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	2	2	-	-	-	-	-	-	-
CO5	-	-	-	-	-	2	-	-	-	2	-	2	-	-
CO6	-	-	-	-	-	1	-	3	2	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ Reference book:

1. Human Values and Ethics in Workplace, United Nations Settlement Program, 2006. (http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
2. Ethics for Everyone, Arthur Dorbin, 2009. (<http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf>) .
3. Values and Ethics for 21st Century, BBVA. (https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)

❖ Web material:

- www.ethics.org

**B. Tech. (Electronics and Communication Engineering)
Programme**

SYLLABI
(Semester – 5)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC 354: ELECTROMAGNETICS & ANTENNA TECHNOLOGY

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Review of Vector Algebra and Vector Calculus	06
2.	Electrostatics	08
3.	Time-Varying Fields and Maxwell's Equations	04
4.	Basic Antenna Concepts	06
5.	Point Sources and Arrays of Point Sources	07
6.	Short Electric Dipole and Thin Linear Antennas	09
7.	Microstrip Antennas	04
8.	Slot, Complementary and Horn Antennas	05
9.	Traveling Wave and Broadband Antennas	05
10.	Reflector and Frequency Independent Antennas	04
11.	Contemporary Antennas	02
Total Hours (Theory):		60
Total Hours (Lab):		30
Total Hours:		90

C. Detailed Syllabus:

1.	Review of Vector Algebra and Vector Calculus (6 Marks)	06 Hours	10%
1.1	Scalars and Vectors, Dot and Cross Products		
1.2	3-D Coordinate Systems – Cartesian, Cylindrical and Spherical and Co-ordinate systems		
1.3	Co-ordinate systems Conversions		
2.	Electrostatics (10 Marks)	08 Hours	13%
2.1	Coulomb's Law, Electric Field Intensity, Field due to Different Charge Distributions		
2.2	Electric Flux Density, Gauss's Law for Electric and Magnetic field, Divergence		
2.3	Current Density, Ampere's Circuital law, Curl		
3.	Time-Varying Fields and Maxwell's Equations (5 Marks)	04 Hours	7%
3.1	Faraday's law, Displacement Current		
3.2	Magnetic flux, Magnetic flux density		
3.3	Maxwell's Equations in Point and Integral forms for time-varying fields		
3.4	Vector Magnetic Potentials and the Retarded Potentials		
4.	Basic Antenna Concepts (7 Marks)	06 Hours	10%
4.1	Radiation Mechanism, Fundamental Parameters		
4.2	Transmission Formula, Sources of Radiation and Antenna Measurement		
4.3	Radio Wave Propagation		
5.	Point Sources and Arrays of Point Sources (8 Marks)	07 Hours	11%
5.1	Introduction to Point Sources, Arrays of Two Isotropic Point Sources, Non-isotropic Point Sources, Principle of Pattern Multiplication		
5.2	Broad Side and End Fire Arrays		
6.	Short Electric Dipole and Thin Linear Antennas (10 Marks)	09 Hours	15%
6.1	Short Electric Dipole: E and H Components, Radiation Resistance		
6.2	Linear Antenna ($\lambda/2$): E and H Components, Radiation Resistance		
7.	Microstrip Antennas (5 Marks)	04 Hours	7%
7.1	Introduction, Feed Methods, Design Examples		
8.	Slot, Complementary and Horn Antennas (6 Marks)	05 Hours	8%
8.1	Slot Antenna: Principle, Radiation Pattern		

8.2	Complementary Antennas		
8.3	Horn Antenna: Types of Rectangular and Circular Horn Antennas		
9.	Traveling Wave & Broadband Antennas (6 Marks)	05 Hours	8%
9.1	Traveling Wave Antennas: V Antenna and Rhombic Antenna		
9.2	Broadband Antenna: Helical Antenna		
10.	Reflector and Frequency Independent Antennas (5 Marks)	04 Hours	7%
10.1	Reflector Antenna: Paraboloidal, Radiation Pattern, Feed Methods		
10.2	Frequency Independent Antennas: Log Periodic and Spiral Antennas		
11.	Contemporary Antennas (2 Marks)	02 Hours	4%
11.1	Introduction to Smart Antennas		
11.2	Introduction to 5G MIMO Antennas		

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the coordinate systems and their applications to solve the real-time multidimensional analytical problems
CO2	Apply Maxwell's equations and their relevant applications for the solution of diverse engineering problems
CO3	Understand the requirements of antennas, antenna parameters, measurement, and its various applications.
CO4	Solve mathematical equations for various fields of thin linear antennas.
CO5	Design, analyze, and model various antennas and antenna arrays.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	-	-	-	-		-	-	-	3	-
CO2	3	3	3	3	-	-	-	-	-	-	-	3
CO3	3	3	3	3	3	-	3	-	2	-	3	-
CO4	3	2	-	3	-	-	3	-	-	2	-	2
CO5	3	3	3	3	3	2	3	-	2	3	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Materials:

❖ **Text Books:**

1. W. H. Hayt & J. A. Buck, “Engineering Electromagnetics”, McGraw Hill, 7th Edition
2. J. D. Krauss, “Antennas and Wave Propagation”, Tata McGraw Hill, 4th Edition

❖ **Reference Books:**

1. Matthew Sadiku, “Elements of Electromagnetics”, Oxford University, 4th Edition
2. Constantine A. Balanis, “Antenna Theory: Analysis and Design”, Wiley, 4th Edition
3. K. D. Prasad, “Antennas & Wave Propagation”, Satya Prakashan Publications

❖ **Web Materials/ Reading Material:**

1. <http://www.electronics-tutorials.com/antennas/antenna-basics.html>
2. NPTEL Lecture series on the course: Electromagnetic Theory
3. http://wireless.ictp.it/school_2006/lectures/Struzak/RadioPropBasics-ebook.pdf
4. Lab Manuals
5. Hand Outs
6. Assignments
7. Question Bank

❖ **Simulators**

1. High Frequency Structure Simulator (HFSS)
2. CST Studio Suite

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC323: VLSI TECHNOLOGY

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hrs/week	4	0	4	4
Marks	100	0	100	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction and fabrication of MOSFET	05
2.	MOS Transistor	08
3.	MOS Inverters: static characteristic and switching characteristic	16
4.	MOS combinational circuits	06
5.	MOS sequential logic circuits	04
6.	MOS dynamic logic circuits	04
7.	FINFET	02

Total Hours (Theory):45

Total Hours (Lab): 30

Total Hours: 75

Detailed Syllabus:

- | | | | |
|-----------|---|----------------|------------|
| 1. | Introduction and fabrication of MOSFET | 05Hours | 11% |
| 1.1 | VLSI Design Flow, Design hierarchy, Design Methodology, | | |
| 1.2 | nMOS,pMOS,CMOS fabrication process. | | |
| 2. | MOS Transistor | 8Hours | 17% |

2.1	Metal Oxide Semiconductor (MOS) structure, The MOS System under external bias.		
2.2.	Structure & Operation of MOS transistor, MOSFET Current-Voltage characteristics		
2.3	MOSFET scaling & small-geometry effects, MOSFET capacitances		
3.	MOS Inverters: static characteristic and switching characteristic	16 Hours	35%
3.1	Introduction, Resistive load Inverter		
3.2	Inverter with n-type MOSFET load (Enhancement & Depletion type MOSFET load)		
3.3	CMOS Inverter		
3.4	Delay-time definitions, Calculation of Delay times, Inverter design with delay constraints		
3.5	Estimation of Interconnect Parasitic, Calculation of interconnect delay		
3.6	Switching Power Dissipation of CMOS Inverters		
4.	MOS combinational circuits	6Hours	13%
4.1	Introduction, MOS logic circuits with Depletion nMOS Loads		
4.2	CMOS logic circuits, ,		
4.3	Complex logic circuits		
4.4	CMOS Transmission Gates (TGs)		
5.	MOS sequential logic circuits	04 Hours	9%
5.1	Introduction, Behaviour of Bistable elements, The SR latch circuit		
5.2	Clocked latch & Flip-flop circuit, CMOS D-latch & Edge-triggered flip-flop		
6.	MOS dynamic logic circuits	04 Hours	9%
6.1	Principles of pass transistor circuits,		
6.2	Synchronous Dynamic Circuit Techniques		
6.3	CMOS Dynamic Circuit Techniques, Domino Logic		
7.	FINFET	02 Hours	4%
7.1	Limitation of CMOS technology, SOI (Silicon on Insulator)		
7.2	FinFET : structure, output characteristics		

Course Outcomes (CO):

At the end of the course student will be able to:

CO1	Understand physics of MOSFET and hence utilize this in analysis of basic building block of digital integrated circuits.
CO2	Model digital integrated circuits using latest CAD/EDA tools used in industry.
CO3	Identify and analyze the recent trends in VLSI Technology and Design methodologies.
CO4	Interpret & Evaluate dynamic logic concept and hence construct high density and high performance digital circuits design. Understand advanced MOSFET structures.
CO5	Design combinational and sequential circuits using various topologies that realizes myriad digital functions of given specification.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	-	2	-	-	-	2	-	3
CO2	3	3	3	2	2	2	-	-	2	2	-	2
CO3	3	3	3	2	2	2	-	-	2	2	-	2
CO4	3	3	3	2	2	2	-	2	2	2	-	2
CO5	3	3	3	3	3	-	-	2	3	3	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Text Books:

1. Sung-Mo-Kang, Usuf Leblebici ,CMOS digital integrated circuits: Analysis and Design, Tata McGrawhill,2003

❖ Reference Books:

1. Douglas Pucknell, Basic VLSI Design,PHI, 1999
2. Wayne Wolf ,Modern VLSI Design., Person Education, 2001
3. John Uyemura ,Introduction to VLSI circuits and systems, Wiley, 2002

❖ Web materials

1. <http://www.eng.auburn.edu/~strouce/elec4200.html>

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC357: DIGITAL SIGNAL PROCESSING

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction	1
2.	Discrete-Time Signals and Systems	6
3.	The Z- Transform	8
4.	Sampling of Continuous-Time Signals	4
5.	Structures for Discrete Time Systems	6
6.	Filter Design Techniques	4
7.	Discrete-Fourier Transform	6
8.	Computation of Discrete-Fourier Transform	6
9.	Contemporary topics in Signal Processing	4
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

1. Introduction	1 Hour	2%
1.1 Introduction, Block diagram of DSP, Advantage and disadvantage of DSP		
2. Discrete-Time Signals and Systems	6 Hours	13%

2.1	Discrete-Time Signals		
2.2	Discrete-Time Systems		
2.3	Linear Time-Invariant Systems		
2.4	Properties of Linear Time Invariant Systems		
2.5	Representation of sequences by Fourier Transform		
2.6	Fourier Transform Theorems		
3.	The Z- Transform	8 Hours	18%
3.1	Z-Transform		
3.2	Properties of the Region of Convergence for the Z-transform		
3.3	The Inverse Z-transform		
3.4	Z-transform properties		
4.	Sampling of Continuous-Time Signals	5 Hours	16%
4.1	Periodic Sampling		
4.2	Frequency domain representation of sampling		
4.3	Reconstructions of band limited signals from its samples		
5.	Structures for Discrete Time Systems	6 Hours	12%
5.1	Block Diagram representation of Linear Constant-Coefficient Difference equations		
5.2	Basic Structures of IIR Systems & Its Transpose.		
5.3	Transposed Forms		
5.4	Basic Network Structures for FIR Systems		
6.	Filter Design Techniques	4 Hours	09%
6.1	Design of Discrete-Time IIR filters from Continuous-Time filters from Continues –Time Filters		
6.2	Design of FIR filters by Windowing		
7.	The Discrete Fourier Transform	6 Hours	12%
7.1	The Discrete Fourier Series		
7.2	Properties of Discrete Fourier Series		
7.3	The Discrete-Fourier Transform		
7.4	Properties of the Discrete Fourier Transform		
7.5	Linear Convolution using the Discrete Fourier Transform		
8.	Computation of Discrete-Fourier Transform	6 Hours	12%

8.1 Efficient Computation of the Discrete Fourier Transform

8.2 The Goertzel Algorithm

8.3 Decimation-in-Time FFT Algorithms

8.4 Decimation-in-Frequency FFT Algorithms

9. Contemporary topics in Signal Processing

4Hours

9%

9.1 Examples of Signal Generation and Visualization

9.2 Measurement and Feature Extraction

9.3 Applications of Image Processing ,Speech Processing and Video Processing

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the need for Digital Signal processing and its various applications on real time signals.
CO2	Selection of Signal clarity techniques using sampling and quantization in time and frequency domain
CO3	Analyse the discrete time signals using convolution /correlation /transformations processing and with DSP algorithms
CO4	Justify the Discrete Time Fourier Transform techniques to its realization as Discrete Fourier Transform using MATLAB

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	--	-	-	-	-	-	-
CO2	3	3	3	3	3	-	-	-	3	-	-	-
CO3	2	3	3	2	3	-	-	-	3	-	-	-
CO4	2	3	3	3	3	-	-	-	3	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Oppenheim, Schafer ,Discrete Time Signal Processing, Buck Pearson education publication, 2nd Edition, 2003
2. Proakis, Manolakis, Digital Signal Processing: Principles, Algorithm & Application, , PHI, 2003, 3rd Edition

3.Sanjit Mitra ,Digital Signal Processing: A Computer Based approach, McGrawHill

MATLAB's user guide.

❖ **Web Materials/ Reading Material:**

7. NPTEL lecture series

8. Lab Manual

9. Assignments

10. Question Bank

11. http://en.wikipedia.org/wiki/Digital_signal_processing

12. <http://cnx.org/content/col10360/latest/>

❖ **Simulators:**

MATLAB

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC358: MICROCONTROLLER AND APPLICATION

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to the AVR microcontroller and features	06
2	AVR timer programming in C	08
3	AVR interrupt programming in C	07
4	AVR serial port programming in C	05
5	LCD and keyboard in interfacing	07
6	ADC, DAC and sensor interfacing	06
7	Relay, optoisolator and stepper motor interfacing.	06
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

- 1. Introduction to the AVR microcontroller and features** **6 Hours 14%**
 - 1.1 Microcontrollers and embedded processors
 - 1.2 Overview of AVR family
 - 1.3 Pin diagram and architecture of AVR
 - 1.4 Arithmetic and logical operations in C

2. AVR timer programming in C	8 Hours	18%
2.1 Normal and CTC modes of AVR timers		
2.2 AVR timers in c to generate a delay		
2.3 AVR counters in c to generate a delay		
3. AVR interrupt programming in C	7 Hours	15%
3.1 Interrupt Vs polling		
3.2 Interrupt service routine(ISR)		
3.3 All the major interrupts of AVR		
3.4 Enable or disable AVR interrupts		
3.5 Programming of AVR timers using interrupts		
3.6 External hardware interrupts of AVR		
4 AVR serial port programming in C	5 Hours	12%
4.1 Serial data transfer and parallel data transfer-comparison with all necessary aspects		
4.2 Serial communication protocol		
4.3 Synchronous vs. asynchronous communication		
4.4 Half duplex and full duplex data transmission		
4.5 Data transfer rate and bps rate		
4.5 Baud rate of AVR		
4.6 RS232 standard, MAX232 and MAX 233 chips		
5 LCD and keyboard in interfacing	7 Hours	15%
5.1 Introduction of LCD		
5.2 Interface an LCD to AVR		
5.3 Programming of LCD in C		
5.4 Interface 4x4 keypad to AVR using C		
6 ADC, DAC and sensor interfacing	6 Hours	13%
6.1 Interfacing of ADC (Analog to digital convertor) to AVR		
6.2 Interfacing of DAC to AVR		
6.3 Interface temperature sensor to AVR		
7 Relay, optoisolator and stepper motor interfacing	6 Hours	13%
7.1 Basic operation of relay		
7.2 Interface relay with AVR		

- 7.3 Basic operation of optoisolator
- 7.4 Interface optoisolator with AVR
- 7.5 Basic operation of stepper motor
- 7.6 Interface stepper motor with AVR

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the need of various Microcontrollers for various embedded applications.
CO2	Understand the formulation, development, debugging and verification of C language programs.
CO3	Apply assembly or high level language and microcontroller according to situation based problems.
CO4	Design various small scale embedded application using knowledge of various external I/O peripherals and sensors.
CO5	Design various time and event critical small scale embedded application using knowledge of programmable internal timers and interrupts.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	3	1	1	-	-	-	-	-	1	-	1
CO2	3	3	-	2	3	-	-	-	2	-	-	-
CO3	3	3	2	2	-	-	-	-	-	-	-	-
CO4	2	2	3	3	3	1	1	-	2	-	-	2
CO5	2	2	3	3	3	1	1	-	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

4. The AVR microcontroller and Embedded Systems by muhammad Ali Mazidi, SarmadNaimi, SepehrNaimi

❖ Web Materials/ Reading Material:

1. Embedded C programming and the ATMEL AVR by Barnett, cox and o’cull, Thomson
2. Hand Outs
3. Assignments
4. Question Bank

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC360: SUMMER INTERNSHIP-I

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	3	-	3
Marks	-	100	-	

Pre-requisite courses:

Programming Language, Software Engineering.

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Summarize theory and practical concepts.
CO2	Assess interests and abilities In the field of study.
CO3	Perceive career alternatives prior to graduation.
CO4	Build a record of work experience.
CO5	Adapt the skill of Team work.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3	-	-	-	2	-	2	-
CO2	3	3	3	3	3	3	-	3	3	3	3	3
CO3	3	3	3	3	3	3	-	3	3	3	3	3
CO4	3	3	3	3	3	-	-	-	2	-	2	-
CO5	3	3	3	3	3	3	-	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Books, Magazines ,Journals & online course platforms of related topics

❖ Web Materials/ Reading Material:

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <https://www.udemy.com/>
5. <https://www.udacity.com/>
6. <https://nptel.ac.in/course.html>
7. <https://www.futurelearn.com/>

SOFTWARES:

Xilinx Altera
MATLAB/PYTHON
HFSS

HS124.01A: PROFESSIONAL COMMUNICATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- Communicative English

Objectives of the Course:

- To hone and sharpen Communication Skills of students
- To prepare globally and multi-culturally competent communicators and professionally compatible cadre of future professionals
- To equip and empower students to qualify and successfully clear all the phases of selection procedure for on and off campus interviews

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An Introduction to Communication	06
2.	Cross-cultural Communication and Globalization	03
3.	Communication for Career Building	10
4.	Group Dynamics and Soft Skills	05
5.	Effective Presentation Strategies	04
6.	Contemporary Issues in Communication and Soft Skills	02
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

- An Introduction to Communication 06 Hours 20%
Basics of Communication: Origin, Concept, Process, Levels,
Principles and Barriers; Applications of Communication;

	Rhetoric in Professional Communication; Importance of Ethos, Logos, and Pathos in Communication		
2.	Cross-cultural Communication and Globalization Basic Concepts: Culture, Globalization and Cross-cultural Communication; Social and People Skills; Communicating with People of Different Cultures; Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them; Persuasive Communication	03 Hours	10%
3.	Communication for Career Building Cover Letters and Resume; E-mail and Report; Types of Resume; Concept and Rationale of Group Discussion Skills and Aspects assessed in Group Discussion; Concept and Rationale of Personal Interview; Types of Personal Interview; Writing Statement of Purpose	10 Hours	33%
4.	Group Dynamics and Soft Skills An Introduction to Group Dynamics and Soft Skills; Groups and their Structures; Roles and Functions of Members in Groups; Conflict Management; Aptitude and Attitude; Various Intelligences; Developing an Open Mindset	05 Hours	17%
5.	Effective Presentation Strategies Designing Appealing Presentation; Audience Analysis and Supporting Material; Presentation Mechanics and Presentation Process; Managing Yourself during Q and A Session; Fundamentals of Persuasion	04 Hours	14%
6.	Contemporary Issues in Communication and Soft Skills Trends and Practices in Communication, Case Studies	02 Hours	06%

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Gain thorough understanding and proficiency in various Professional Communication Skills.
CO2	Develop awareness and competence in cross-cultural communication in their personal, academic and professional environments.
CO3	Develop business writing and presentation skills to succeed in career.
CO4	Develop soft skills to stand out and take their career to the next level.
CO5	Develop various intelligences and open Mindset to function in multi-disciplinary and cross-cultural work environment.

CO6	Practice new trends in communication in multiple perspectives at personal, professional, and social level.
-----	--

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	2	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	3	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. Koneru, A. Professional Communication, Tata McGraw Hill Education Private Limited
2. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
3. Raman, M & Singh, P. Business Communication, Oxford University Press

❖ **Reference book:**

4. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
5. Anandamurugan, A. Placement Interviews – Skills for Success, Tata McGraw Hill Education Private Limited

❖ **Web material:**

1. <https://www.coursera.org/learn/careerdevelopment>
2. <https://www.futurelearn.com/courses/writing-applications>
3. <https://www.futurelearn.com/courses/workplace-englis>

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC314: MODELING USING VERILOG

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	4	4	4
Marks	0	100	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to digital circuit design flow	06
2	Verilog variables, operators and language constructs	06
3	Modeling combinational circuits using Verilog	06
4	Modeling sequential circuits using Verilog	06
5	Verilog test benches and design simulation	06
	Total Hours (Theory):	00
	Total Hours (Lab):	30
	Total Hours:	30

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the need and trends in verification and python programming
CO2	Apply the concepts of Object Oriented Programming, task and functions in creating verification test bench of a digital module.
CO3	Analyze and verify the functionality of digital module by implementing overall verification test bench environment.

CO4	Able to Express different Decision-Making statements ,Control Statements and Interpret Object Oriented programming in Python
CO5	Design Firmware for OS based Hardware

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	0	0	0	0	0	0	0	1	0	1
CO2	2	1	0	0	3	0	0	0	2	2	0	1
CO3	3	3	0	2	3	0	1	0	3	3	0	2
CO4	1	3	0	2	3	0	0	0	3	3	0	2
CO5	3	1	3	2	3	0	0	0	3	3	0	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

❖ Web Materials/ Reading Material:

1. Hand Outs
2. Assignments
3. Question Bank

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC385: MOBILE APPLICATION DEVELOPMENT

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	50	50	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to MIT App Inventor	02
2	Introduction to Android	04
3	User Interface Using Views and View Groups	06
4	Intents and Fragments in Android	08
5	Database Connectivity	06
6	Publishing and Distributing Android Application	04
	Total Hours (Theory):	00
	Total Hours (Lab):	30
	Total Hours:	30

C. Detailed Syllabus:

Sr. No.	Chapter detail	Hours	Weightage
1.	Introduction to MIT App Inventor Getting Started with MIT App Inventor Elements of MIT App Inventor Developing and Executing the first Mobile Application using MIT App Inventor	02	15%
2.	Introduction to Android Android Architecture Downloading and Installing Android Exploring the Development Environment Developing and Executing the first Android Application	04	15%
3.	User Interface Using Views and View Groups Working with Views- Text Edit Text, Button, Radio Button, Check box, Image Button, Toggle Button, Rating Bar Working with View Groups- Linear Layout, Relative Layout, Constraint Layout, Scroll View Table, Frame, Table with Action Bar Binding Data with the Adapter View Class List View, Spinner, Gallery View Creating Menus & Dialogs	06	15%
4.	Intents and Fragments in Android Intent Objects & Intent Filters, Linking the Activities Using Intent, Obtaining Results from Intent, Passing Data Using an Intent Object, Fragments- Fragment Implementation, Finding Fragments, Adding, Removing, and Replacing Fragments	08	20%
5.	Database Connectivity Introducing the Data Storage Options,	06	20%

	Data storage using Internal storage & External storage, Data storage using SQLite Database Working with content Provider		
6.	Publishing and Distributing Android Application Signing the Android Application Versioning the Android Application Publishing the Android Application	04	15%

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand Android architecture, activities and their life cycle.
CO2	Able to select View Groups comprising layouts and Views in application.
CO3	Able to manage data binding, user interface events, maps
CO4	Able to Publish and distribute Android Application

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	2	3	1	-	-	-	-	-	-
CO2	2	2	3	2	3	1	-	-	-	-	1	-
CO3	3	2	3	2	3	1	-	-	-	-	1	-
CO4	3	2	3	2	3	1	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

5. Android Application Development Black Book by Pradeep Kothari, DreamTech
6. Beginning Android Application Development by Wei Meng Lee, Wrox

❖ Web Materials/ Reading Material:

5. <https://developer.android.com/>
6. Hand Outs
7. Assignments
8. Question Bank

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC386: INTRODUCTION TO SCRIPTING LANGUAGE

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	50	50	

D. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Python Basic Syntax	05
2	Conditional Statements	05
3	Python Modules	10
4	Classes & Object-Oriented Programming	10
	Total Hours (Theory):	00
	Total Hours (Lab):	30
	Total Hours:	30

E. Detailed Syllabus:

1.	Python Basic Syntax	05 Hrs	10 %
1.1	Variable & Python Data Types		
1.2	Number Data Type		
2.	Python Conditional Statement	05 Hrs	10 %
2.1	a) If else condition		
2.2	b) For Loop		
2.3	c) While Loop		
3.	Python Modules	10 Hrs	15 %

- 3.1 Function
- 3.2 Function Parameters
- 3.3 Packages
- 3.4 Module
- 4. Class & Object-Oriented Programming 10 Hrs 15 %
- 4.1 Class & Object
- 4.2 Inheritance
- 4.3 Operator Overloading
- 4.4 Polymorphism

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the need and trends of Python programming
CO2	Understand the fundamentals of data structures, flow control mechanisms, regular expressions, and subroutines
CO3	Create functional Python Scripts for text processing application.
CO4	Able to Express different Decision-Making statements ,Control Statements and Interpret Object Oriented programming in Python

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	0	2	0	0	0	2	2	1	1
CO2	2	1	1	0	3	0	0	0	2	2	2	3
CO3	2	2	1	2	3	0	0	0	3	3	2	2
CO4	1	3	0	2	3	0	0	0	3	3	0	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

Python’s Companion by Joe Thompson

Python For Everyone by Cay Horstmann & Rance D. Necaise Willy Publication

❖ Web Materials/ Reading Material:

1. https://www.tutorialspoint.com/perl/perl_introduction.htm
2. Hand Outs
3. Assignments

**B. Tech. (Electronics and Communication Engineering)
Programme**

SYLLABI
(Semester – 6)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC319: DESIGN, TESTING AND VERIFICATION

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

Pre-requisite Course:

- Digital Electronics

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Programmable logic devices & System On Chip	10
2	Finite State Machines and State machine design	11
3	Stuck at faults	06
4	Design for Testability	06
5	Hardware Description Language	12
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

1	Introduction to Programmable logic devices & System On Chip	10 Hours	22%
1.1	Programmable logic versus Discrete logic		
1.2.	Evolution: PROM, PLA, PAL ,Introduction to memories		
1.3.	Classification of FPGA, Basic architecture of CPLD and FPGA, Xilinx's CPLD and FPGA architecture		
1.4	ASIC Design flow & SOC		
2	Finite State Machines and State machine design and SM chart	11 Hours	24%
2.1	Mealy and Moore Machine, Design of sequence detector		
2.2	Design of Elevator, Traffic light controller and vending machine ,digital combination lock with FSM		
3	Stuck at faults	06 Hours	14%
3.1	Introduction, Test process and Test economics		
3.2	Defects, Errors, Faults and Fault Modeling mainly stuck at fault modeling)		
3.3	Fault Equivalence, Fault Dominance		
4	Design for testability	06 Hours	14%
4.1	Functional vs. Structural Testing, scan methods		
4.2	Scan insertion, Scan chain		
4.3	DFT examples		
5	Hardware Description Languages (Verilog)	12 Hours	26%
5.1	Fundamentals of Verilog, Overview of Digital Design with Verilog HDL		
5.2	Hierarchical Modeling Concepts: Top-down and bottom-up design methodology, Modules and Port		
5.3	Gate-Level Modeling, Dataflow Modeling, Behavioral Modeling, Tasks and Functions, RTL Description of Simple Machine and Design from RTL description		

Self-Study:

The self-study contents will be declared at the commencement of semester. Around 10% of the questions will be asked from self-study contents.

Course Outcome (Cos):**At the end of the course student will be able to:**

CO1	Understand the concepts of digital electronics and programmable logic devices
CO2	Apply the concept of state machines to design digital systems and solve real time problems
CO3	Discriminate testing and verification concepts for the system
CO4	Create digital system models using Hardware Descriptive Language in Electronic Design Automation tools.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	-	-	-	-	-	2
CO2	2	3	3	-	-	-	-	3	3	3	3	-
CO3	3	3	2	-	-	-	-	-	-	-	-	-
CO4	2	2	3	2	3	-	3	-	3	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Instructional Method and Pedagogy:

1. At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
2. Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
3. Attendance is compulsory in lectures and laboratory which carries a 5% component of the overall evaluation.
4. Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
5. Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5%.
6. Surprise tests will be conducted which carries 5% component of the overall evaluation.
7. Minimum 7 tutorials which include solution of minimum 5 numerical under each head will be carried out

in laboratory.

Recommended Study Material:

❖ Text Books:

1. Charles H. Roth, Digital System Design Using VHDL, Thomson, 2002
2. J.Bhasker, A Verilog HDL Primer, Third Edition
3. Essential of electronic testing, Michael Bushnell, Vishwani D Agrawal. Kluwer academic publishers new york, boston, dordrecht, london, moscow

❖ Reference Books:

1. Ian Grout Elsevier, Digital Systems Design with FPGAs and CPLDs
2. William I. Fletcher, An Engineering Approach To Digital Design, PHI
3. Samir Palnitkar, Verilog HDL : A Guide to Digital Design and Synthesis, Prentice Hall

❖ Web Material:

1. <http://www.eng.auburn.edu/~strouce/elec4200.html>

❖ Software:

Xilinx Altera

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC362: DIGITAL COMMUNICATION

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Probability Theory & Random processes	12
2.	Information Theory	10
3.	Formatting A Base Band Modulation	07
4.	Principle of digital data transmission	08
5.	Error Correction Codes	02
6.	Digital Modulation Techniques	06
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

- 1. Probability Theory & Random Processes** **12 Hours 25%**
 - 1.1 Introduction to digital communication, Probability, Conditional Probability Of Independent Events
 - 1.2 Relation between probability and probability density ,CDF, PDF, Gaussian PDF, Threshold Detection

1.3	Random Variable, Variance of a random variable, co relation between random variables, Mean and variance of sum of random variables		
1.4	Statistical Average, Chebyshev In Equality, Raleigh Probability Density, The Central Limit Theorem, correlation		
1.5	random processes, PSD of random processes(along with examples)		
1.6	Multiple random processes, band pass random processes		
2.	Information Theory	10 Hours	22%
2.1	Concept & Measure Of Information, entropy.		
2.2.	Error Free Communication Over A Noisy Channel, The Channel Capacity Of A Discrete Memory Less Channel		
2.3	Channel Capacity Of A Continuous Channel		
2.4	Practical Communication Systems In Light Of Shannon's Equation		
3.	Formatting A Base Band Modulation	7 Hours	18%
3.1	Base Band System, the Sampling Theorem, Aliasing		
3.2	Pulse Code Modulation, DPCM, Uniform And Non-Uniform Quantization, Quantization Error In PCM, Non- Uniform Quantization		
3.3	Delta Modulation, Adaptive Delta Modulation, SNR Calculation		
3.4	Phase Modulation, QAM, SNR Calculation		
4.	Principle of digital data Transmission	8 Hours	19%
4.1	Line Coding, PSD Of On/Off Signal, Bipolar Signal, Duo Binary Signal, Pulse Shaping		
4.2	Nyquist First And Second Criterion For Zero ISI, Regenerative Repeaters	10 Hours	16%
4.3	Detection Error Probability, M-ary System		
4.4	Scrambling, Digital Carrier System, Digital Multiplexing		
5.	Error Correction Codes	2 Hours	5%
5.1	Introduction and basics		
6.	Digital Modulation Techniques	6 Hours	11%
6.1	DEPSK, DPSK		
6.2	BPSK, QPSK		

- 6.3 QASK, MSK, M-ary-FSK, M-ary-PSK
- 6.4 BFSK, Comparison Of Various Digital Modulation Techniques
- 6.5 Coherent And Non Coherent Detection Of ASK, FSK, PSK, DPSK

Course Outcomes (CO)

At the end of the course student will be able to:

CO1	Explain the fundamental concept and need of digital communication system for various applications.
CO2	Understand and Apply probability theory and random process by providing applications to communication problems.
CO3	Apply basic principles of information theory to achieve error free communication over noise channel, uniquely decodable code for source encoding and basic principle of practical communication systems.
CO4	Determine the basic principles of transmission of digital data and Pulse shaping techniques.
CO5	Discuss the basic principles of M-ary wired and wireless digital modulation schemes for various applications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	3	-	-	2	-	-	-	2
CO2	3	3	3	3	-	-	-	-	3	-	-	2
CO3	3	3	3	3	3	-	-	2	-	-	-	2
CO4	3	3	3	3	2	-	-	2	-	-	-	2
CO5	3	2	2	2	2	-	-	-	-	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Taub & Schilling, Principle of communication system, TMH Publication
2. B. Carlson, Communication system, TMH Publication
3. John G. Proakis, Digital Communications, TMH Publication
4. Simon Haykin, Digital and analog communication system, Willey Publication

❖ Simulators:

MATLAB

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC368: INTRODUCTION TO EMBEDDED LINUX

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	2
Marks	0	50	50	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Embedded Linux	2
2.	Boot Loader	3
3.	Kernel Internals & Construction	5
4.	File systems	6
5.	Embedded Linux Device Drivers	8
6.	Advance driver concepts	6
	Total Hours (Lab):	30
	Total Hours:	30

Detailed Syllabus:

- | | | |
|--|----------------|-------------|
| 1. Introduction of Embedded Linux | 2 Hours | 07 % |
| 1.1 Why Embedded Linux? | | |
| 1.2 Anatomy of Embedded Linux | | |
| 1.3 Embedded Linux Distribution | | |
| 1.4 Embedded Linux Vs Desktop Linux | | |
| 1.5 User Space Vs Kernel Space | | |

1.6	GNU Cross Platform Tool chain		
2.	Boot loader	3 Hrs	07 %
2.1	Role of Boot loaders		
2.2	Boot loader Challenges		
2.3	Porting U-Boot		
3	Kernel Internals & Construction	5 Hrs	33 %
3.1	Linux Kernel Construction		
3.2	Kernel Build System		
3.3	Obtaining Linux Kernel		
3.4	Composite Kernel Image		
3.5	Kernel Command line Processing		
3.6	Kernel Init Process		
3.7	Initial RAM Disk		
3.8	Kernel Debugging Techniques		
3.9	Real Time Kernel Patch		
3.10	Busy box Configuration & Operation		
4	File systems	06 Hrs	13 %
4.1	Root File System		
4.2	Different type file Systems		
4.3	Difference between initrd and file system		
4.4	obtain a file system for any architecture		
4.5	file system generation for ARM board using DE bootstrap		
5	Embedded Linux Device Drivers	8 Hrs.	22 %
5.1	Types of Device Drivers		
5.2	Linux Kernel Modules		
5.3	Character Device Drivers		
5.4	Driver Debugging Techniques		
6	Advance Driver concepts	6 Hrs	18 %
6.1	Peripheral specific Linux driver sub-systems		
6.2	Device Tree Concept		

- 6.3 Walk-thru of Linux UART & I2C driver subsystem in Beagle bone Black source code
- 6.4 Linux Kernel Porting – Why and How

Course Outcomes (CO):

At the end of the course student will be able to:

CO1	Understand about Linux operating system.
CO2	Understand the Linux system structure and the mechanism of Linux system services
CO3	Analyse Fully comprehend advanced knowledge in Linux such as how to use kernel debugging tools
CO4	Develop C Programing skills and learn cross compilation
CO5	Develop Device Driver Capabilities.
CO6	Design Configuration, porting and installation process of Hardware on Linux Operating System

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	3	-	3	-	-	1	-	1	-	-
CO2	-	-	1	1	3	-	-	2	-	-	-	-
CO3	3	-	2	1	3	-	-	-	2	-	-	-
CO4	3	-	3	3	3	-	-	-	2	2	-	3
CO5	3	-	2	3	3	-	-	-	2	1	-	2
CO6	1	1	-	-	2	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

Embedded Linux Primer by Christopher Hallinan, Prantice Hall

Embedded Linux System Design & Development by P. Raghvan, Amol Lad, Sriram Neelakandan

Linux Device Driver, Jonathan Corbet, Alessandro Rubini, and Greg Kroah-Hartman, O'Reilly

❖ Web Materials/ Reading Material:

Introduction to Linux A Hands on Guide by Machtelt Garreles

Building Embedded Linux System By Karim Yaghmour, O' Reilly

Essential Linux Device Drivers By Sreekrishnan Venkateswaran

Hand Outs

Assignments

Question Bank

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC364: EMBEDDED SYSTEMS

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Embedded Systems	03
2	ARM Processors	15
3	Real Time Operating System	12
4	Wireless Devices and Communication Protocols	12
5	Embedded Product Development Life Cycle	04
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

1	Introduction to Embedded Systems	03 Hours	7%
1.1	Embedded systems and its characteristics with examples		
1.2	Embedded System's design metrics & challenges, Processor Technology, IC Technology, Design Technology		
1.3	Importance of Design Metrics, Example of Embedded systems Working Models		
2	ARM Processors	15 Hours	33%

2.1	Introduction of Single purpose processors ,General purpose processors, Application specific instruction set processors (ASIPs), CISC vs. RISC		
2.2	ARM architecture Inheritance, ARM programmer's model, Memory organization, ARM development tools, ARM data flow model		
2.3	ARM processor modes, ARM instruction set, Thumb instruction set		
2.4	3-pipeline architecture, 5-stage pipeline architecture, Pipeline Hazards		
2.5	STM32 Architecture		
2.6	STM32 GPIO Programming, STM32 UART Programming, STM32 Interrupt Programming		
3	Real Time Operating System	12 Hours	26%
3.1	GPOS Vs RTOS, Scheduling Algorithms, Task Life Cycle, Selection Criteria for RTOS		
3.2	Task Management		
3.3	Queue Management		
3.4	Interrupt Management		
3.5	Resource Management		
3.6	Live lock, Dead lock, Priority Inversion, Inter Task Communication Issues & Prevention		
4	Wireless Devices and Communication Protocols	12 Hours	26%
4.1	Serial bus communication protocols: RS232C, RS485, CAN bus, Modbus, USB bus		
4.2	Parallel bus device protocols: ISA bus, PCI and PCI/x bus, AMBA bus: APB, AHB & AXI		
4.3	Wireless devices: Irda, Bluetooth, GSM, RFID, Zigbee		
5	Embedded Product Development Life Cycle	04 Hours	8%
5.1	Different Phases of EDLC		
5.2	Different Types of EDLC Models		

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the fundamental skills knowledge of embedded system, different examples, its characteristics
CO2	Understand the different processors architecture and able to do a programming for ARM 7 Processor
CO3	Understand operating system and do basic programming of real time operating system
CO4	Identify different networks which are used in embedded system design and interface various peripheral devices
CO5	Analyze the systems requirements to meet the specifications

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	1	-	-	-	-	-	1	-	1
CO2	2	3	3	2	3	-	-	-	2	-	-	-
CO3	3	3	3	2	-	-	-	-	-	-	-	-
CO4	2	3	3	3	3	1	1	-	2	-	-	-
CO5	3	3	2	3	2	1	1	-	2	-	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Raj Kamal, Embedded Systems Architecture, programming and design (Tata –MGHill)
2. Steve furber, ARM System-on-chip Architecture, Pearson

❖ Web Materials/ Reading Material:

1. David E. Simon, An Embedded Software Primer, Addison-Wesely.
2. Frank vahid& Tony D.Givargis, System Design: A unified Hardware/Software introduction, Addison Wesley 2002
3. Wayne Wolf, Computers as components Principles of embedded computing
4. Tammy Noergaard, Embedded Systems Architecture, Elsevier

5. Anderw N. SLOSS, Dominic SYMES, Chirs WRIGHT, ARM System Developer's Guide, Elsevier.
6. Daniel W.Lewis, Fundamentals of Embedded Software: Where C and Assembly Meet Prentice Hall
7. Embedded C programming and the ATMEL AVR by Barnett, cox and o'cull, Thomson
8. Hand Outs
9. Assignments
10. Question Bank

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC322: SYSTEM VERILOG PROGRAMMING

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	4	4	4
Marks	0	100	100	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to System Verilog	03
2	Data types in System Verilog.	05
3	System Verilog Control flow.	07
4	System Verilog Processes.	05
5	System Verilog Communication	07
6	System Verilog Interface	06
7	System Verilog Class (OOPS)	07
8	Python Basic Syntax	05
9	Conditional Statements	05
10	Python Modules	05
11	Classes & Object-Oriented Programming	05
	Total Hours (Theory):	00
	Total Hours (Lab):	60
	Total Hours:	60

Detailed Syllabus:

1.	Introduction to System Verilog	3 Hours	05 %
1.1	Limitation of Verilog		
1.2	Verification		
1.3	SV Attributes for verification		
2.	Datatypes in System Verilog.	5 Hrs	10 %
2.1	String		
2.2	Dynamic Arrays		
2.3	Associative arrays		
2.4	Queues		
3	System Verilog Control flow.	07 Hrs	10 %
3.1	If-else		
3.2	Tasks		
3.3	Functions		
3.4	Loops, Break and continue		
4	System Verilog Processes.	05 Hrs	10 %
4.1	Fork join		
4.2	Disable fork join		
4.3	Wait fork		
5	System Verilog Communication	07 Hrs.	5 %
5.1	Interprocess communication		
5.2	Semaphores		
5.3	Mailbox		
6	System Verilog Interface	6 Hrs	10 %
7	System Verilog Class (OOPS)	07 Hrs	10%
8	Python Basic Syntax	07 Hrs	10%
8.1	Variable & Python Data Types		
8.2	Number Data Type		
9	Python Conditional Statement	5 Hrs	10%
9.1	If else condition		
9.2	For Loop		
9.3	While Loop		

10	Python Modules	5 Hrs	10%
10.1	Function		
10.2	Function Parameters		
10.3	Packages		
10.4	Module		
11	Class & Object-Oriented Programming	5 Hrs	10%
11.1	Class & Object		
11.2	Inheritance		
11.3	Operator Overloading		
11.4	Polymorphism		

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand the need and trends in verification and python programming
CO2	Apply the concepts of Object Oriented Programming, task and functions in creating verification test bench of a digital module.
CO3	Analyze and verify the functionality of digital module by implementing overall verification test bench environment.
CO4	Able to Express different Decision-Making statements ,Control Statements and Interpret Object Oriented programming in Python
CO5	Design Firmware for OS based Hardware

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	0	0	0	0	0	0	0	1	0	1
CO2	2	1	0	0	3	0	0	0	2	2	0	1
CO3	3	3	0	2	3	0	1	0	3	3	0	2
CO4	1	3	0	2	3	0	0	0	3	3	0	2
CO5	3	1	3	2	3	0	0	0	3	3	0	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

9. System Verilog for Verification: A Guide to Learning the Testbench Language Features By Chris Spear
10. A Practical Guide for System Verilog Assertions by Meyyappan Ramanathan and Srikanth Vijayaraghavan
11. Python's Companion by Joe Thompson
12. Python For Everyone by Cay Horstmann & Rance D. Necaise Willy Publication

❖ Web Materials/ Reading Material:

13. SystemVerilog 3.1a Language Reference Manual
(http://www.ece.uah.edu/~gaede/cpe526/SystemVerilog_3.1a.pdf)
14. Hand Outs
15. Assignments
16. Question Bank

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC321: PROGRAMMING IN JAVA

Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Fundamental of Object-Oriented Programming	4
2.	Class Fundamentals	7
3.	Array & String Handling	6
4.	Inheritance, Interfaces & Packages	7
5.	Exceptions Handling	6
6.	Multithreaded Programming	7
7.	GUI Programming & Lambdas and Streams	8
	Total hours (Theory):	45
	Total hours (Lab):	30
	Total hours:	75

Detailed Syllabus:

- Fundamental of Object-Oriented Programming** **04 Hours 08 %**
 History of Java, Basic overview of java, Bytecode, JVM, Buzz-words, Application and applets, Constants, Variables & Data Types, Comments, Operators, Control Flow
- Class Fundamentals** **07 Hours 16 %**
 General form of class, Creating class Overloading methods, Constructor, Declaring Object, Returning objects, using objects as parameters, Assigning object reference variables, Introducing Access

control, Understanding static, Introducing final, The finalize() method, The this keyword, Garbage collection

- | | |
|---|-------------------------|
| 3. Array & String Handling | 06 Hours 13 % |
| Array basics, String Array, String class, StringBuffer class, String Tokenizer Class and Object Class | |
| 4. Inheritance, Interfaces & Packages | 07 Hours 16 % |
| Inheritance: Using super creating multilevel Hierarchy, method overriding, Dynamic method dispatch, abstract classes, using final with Inheritance, Using Package: Defining package, finding package and CLASSPATH, Access protection, importing package, Interface: Defining Interface, Implementing Interface, Variables in Interface | |
| 5. Exceptions Handling | 06 Hours 13 % |
| Exception types, Try ...Catch...Finally, Throw, Throws, creating your own exception subclasses | |
| 6. Multithreaded Programming | 07 Hours 16 % |
| Life cycle of thread, thread methods, thread priority, thread exceptions, Implementing Runnable interface, Synchronization | |
| 7. GUI Programming & Lambdas and Streams | 08 Hours 18 % |
| Introduction to Annotation, Byte streams and character streams, Wrapper classes , Why Lambda Expression, Lambda Expression Syntax, Where to use lambda expression, Adopting Patterns like matching, finding and filtering, Swing overview ,Swing component classes: | |

Course Outcomes (CO):

After learning the course, students will able to

CO1	Implement Object Oriented programming concept using basic syntaxes of control Structures, strings, and function for developing skills of logic building activity.
CO2	Use of a variety of basic control structures including selection and repetition; classes and objects in a tiered architecture (user interface, controller, and application logic layers)
CO3	Demonstrates how to achieve reusability using inheritance, interfaces, and packages and describes faster application development that can be achieved.

CO4	Demonstrate understanding and use of different exception handling mechanisms and concepts of multithreading for robust faster and efficient application development.
CO5	Identify and describe common abstract user interface components to design GUI in Java using Swing along with a response to events.
CO6	Identify, Design & develop complex Graphical user interfaces using principal Java Swing classes based on MVC architecture

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	3	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	3	2	1	1	-	-	-	-	-	-	1	3
CO3	-	2	2	3	2	-	-	-	-	-	1	1	-	-
CO4	1	1	1	1	3	-	-	-	-	-	2	2	3	-
CO5	1	1	2	3	3	-	-	1	-	-	1	1	2	2
CO6	1	1	1	2	3	1	-	1	-	-	2	1	3	3

F. Recommended Study Material:

❖ Text Books:

1. Java: The Complete Reference, Eleventh Edition by Herbert Schildt, Oracle Press

❖ Reference Books:

1. Java: A Beginner's Guide, Eighth Edition 8th Edition by Herbert Schildt, Oracle Press
2. Head First Java: A Brain-Friendly Guide 2nd Edition by Kathy Sierra, Bert Bates, O'Reilly
3. OCP Oracle Certified Professional Java SE 11 Programmer I Study Guide: Exam 1Z0-815 1st Edition by Jeanne Boyarsky, Scott Selikoff

❖ Web Materials:

1. <https://docs.oracle.com/javase/tutorial/tutorialLearningPaths.html>
2. <http://openjdk.java.net/projects/jigsaw/>
3. <https://docs.oracle.com/en/java/javase/14/docs/api/index.html>

❖ Software

1. <https://www.oracle.com/java/technologies/javase-downloads.html>
2. <https://netbeans.apache.org/download/index.html>
3. <https://download.eclipse.org/eclipse/downloads/>

HS132.02 A: CONTRIBUTORY PERSONALITY DEVELOPMENT

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- Communication and Soft Skills

Objectives of the Course:

- Become familiar with basic concept of personality and personality development
- Understand personality development theories and strategies
- Evaluate one's personality and inculcate traits of an assertive personality
- Develop an assertive personality
- Develop life skills and required management traits
- Enhance contributory personality for academic and career success

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Concept of Personality	06
2.	Soft Skills and Personality Development	08
3.	Developing Contributory Personality	06
4.	Life skills and Personality Development	06
5.	Contemporary Issues in CPD	04
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

- | | | | |
|----|--|----------|-----|
| 1. | Concept of Personality
Meaning of Personality, Types of Personality, Factors contributing to Personality, Personality Traits, Personality Profiling | 06 Hours | 20% |
| 2. | Soft Skills and Personality Development | 08 Hours | 26% |

Positive Thinking and Mind Set, Leadership, Assertiveness and Negotiation Skills, Self-Management, Interpersonal Skills, Being a Team Player

3. Developing Contributory Personality 06 Hours 20%
Concept of Contributory Personality, Characteristics of a Contributor, The Contributor's Vision of Success & Career, The Scope of Contribution in a field, Embarking on the Journey to Contributor ship, Developing Contributor Personality, Reviewing Some Contributors Personalities
4. Life skills and Personality Development 06 Hours 20%
Concept of life skills, Self-awareness, Empathy, Decision Making, Problem Solving
5. Contemporary Issues in CPD 04 Hours 14%
Contemporary Trends and Practices in Contributory Personality Development, Case Study & Presentations

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify one's individual personality strengths and challenges.
CO2	Develop more assertive and optimist attitude towards work and life.
CO3	Develop quintessential soft skills to groom one's personality.
CO4	Identify traits of contributor personality.
CO5	Contribute to self, society, nation, and globe.
CO6	Develop skills of global citizenship to perform societal responsibilities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	2	-	2	2	-	-	-	-
CO6	-	-	-	-	-	2		-	-	2	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

4. Personality Development & Soft Skills, Oxford University Press
5. Soft Skills, Bookboon
6. Personality Development, Swami Vivekananda; Advaita Ashrama

❖ **Reference book:**

6. Contributor Personality Program Workbook (Volume 1,2),
7. Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd

❖ **Web material:**

4. <https://www.coursera.org/learn/wharton-succcess>
5. <https://www.coursera.org/learn/personality-types-at-work>
6. <https://www.coursera.org/learn/self-awareness>

**B. Tech. (Electronics and Communication Engineering)
Programme**

SYLLABI
(Semester – 7)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC451: DATA COMMUNICATION AND NETWORKING

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Basics of Data Communication and Networking	7
2.	Physical Layer	8
3.	Data Link Layer	10
4.	Network Layer	10
5.	Transport Layer	10
6.	Application Layer	7
7.	Network Security	8
	Total Hours (Theory):	60
	Total Hours (Lab):	30
	Total Hours:	90

Detailed Syllabus:

- | | | |
|---|----------------|------------|
| 1. Basics of Data Communication and Networking | 7 Hours | 12% |
| 1.1 Data communication, Networks, Internet | | |
| 1.2 Protocols and standards | | |
| 1.3 Layering of Models, OSI model, Internet model. | | |
| 2. Physical Layer | 8 Hours | 13% |

2.1	Signals, digital and analog transmission, multiplexing		
2.2	Transmission media, circuit switching and telephone network		
2.3	Digital Subscriber Line (DSL), Cable Modems and SONET		
3.	Data Link Layer	10 Hours	17%
3.1	Error detection and correction		
3.2	Data link control and protocols, Point to Point Protocol		
3.3	Multiple Access Techniques (CSMA, CSMA/CD, CSMA/CA)		
3.4	Ethernet, Wireless LANs(IEEE 802.11, IEEE 802.15)		
3.5	Connecting Devices (Hub, Bridges, Switch, Router, Gateways)		
3.6	Backbone Networks and virtual LANs		
3.7	Frame Relay and ATM		
4	Network Layer	10 Hours	17%
4.1	Packet Switching, Virtual circuits and datagram		
4.2	Static and Dynamic Routing Algorithms (Optimality principle, Static Routing Algorithms, Shortest Path, Flooding, Dynamic routing Algorithms, Distance Vector, Link state routing.)		
4.3	IP Addressing , CIDR & NAT, IP layer protocols (ICMP, ARP, RARP, DHCP, BOOTP,), IPv6		
4.4	Congestion control Algorithms (Principles, policies, Algorithms)		
4.5	QoS-quality of Service (Integrated services & Differentiated Services)		
5.	Transport Layer	10 Hours	17%
5.1	Elements of Transport protocols,		
5.2	Internet protocols: TCP & UDP		
5.3	Introduction to Socket programming and QOS of Network layer and transport layer		
5.4	Congestion control & QoS		
6.	Application Layer	7 Hours	10%
6.1	DNS-Domain Name System		
6.2	E-mail, World Wide Web		
6.3	Introduction to File Transfer Protocol (FTP)		
6.4	Introduction to Multimedia over networks		

6.5 Introduction to VoIP

7 **Network Security**

8 Hours 13%

7.1 Cryptography

7.2 Symmetric key Algorithms, DES, AES

7.3 Public key Algorithms, RSA

7.4 Digital Signatures, Firewall, IPSec

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understanding basics concepts and model of data communication and networking.
CO2	Classify and Compare routing protocols and medium access control protocol for data communication.
CO3	Analyze the layer architecture and protocols for data communication and Networking.
CO4	Identify the importance of quality of services and network security for data communication and networking.
CO5	Elaborate the relevant services of application layer.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3	-	-	-	-	-	-	-
CO2	3	3	3	3	3	-	-	-	2	-	2	1
CO3	3	3	3	3	3	-	-	-	2	-	-	-
CO4	3	3	3	3	3	-	-	-	3	-	-	2
CO5	2	2	2	2	2	-	-	-	2	2	-	

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Data and Communication Networking By Behrouz Forouzan, TMH
2. Computer Networks by A. S. Tanenbaum, PHI
3. Data Networks, by W. Stalling, Pearson Education India

❖ Web Materials/ Reading Material:

11. <http://citengg.blogspot.com/p/behrouz-forouzancomputer-networks4th.html>
12. Lab Manuals
13. Hand Outs
14. Assignments
15. Question Bank

Simulators:

Cisco Packet Tracer, NS2

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC452: RF & MICROWAVE ENGINEERING

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	1	4	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to microwaves	02
2.	Microwave transmission lines	12
3.	Microwave waveguides	10
4.	Microwave components & their s-parameters.	08
5.	Microwave diodes and amplifier	08
6	Phase Locked Loop	05
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

1. Introduction to microwaves	02 hours	3%
1.1 Microwave frequencies, advantages of microwaves, and general applications of microwaves		
2. Microwave transmission lines	12 hours	33%
2.1 Transmission line equations & solutions		
2.2 Reflection and transmission coefficient, standing wave and standing wave ratio		
2.3 line impedance and admittance, smith chart, impedance matching		
3. Microwave waveguides	10 hours	20%
3.1 Rectangular waveguides with propagation modes: TE, TM and TEM		
4. Microwave passive components & their s-parameters.	08 hours	17%
4.1 Introduction of s-parameters		
4.2 Waveguide tees: E plane, H plane and magic tee		
4.3 waveguide corners, bends, twists		
4.4 Directional couples, circulars and isolators.		
5. Microwave diodes and amplifier	08 hours	17%
5.1 Introduction to Microwave diodes: PIN diode, Varactor diode		
5.2 Design of Low noise amplifier		
Performance parameters: Impedance matching, Stability , Noise factor, Gain , IIP3 /OIP3,1 dB compression bandwidth		
6. Phase Locked Loop	05 hours	10%
Introduction, Phase frequency detector, Loop Filter		
Voltage Controlled Oscillator, Pre-scalar, Reference divider		
Performance measurement of PLL (Phase Noise, Lock time, Frequency spurs, stability), Applications		

Course Outcome (CO):

At the end of the course student will be able to

CO1	Understand the fundamentals of microwave frequency bands and their application areas
CO2	Analyse the transmission line with the use of Smith chart to design microstrip and strip lines by evaluating the parameters like VSWR, return loss, insertion loss and impedance matching.
CO3	Evaluate the mode equations of microwave wave guides by designing the different frequency bands waveguide.
CO4	Design and Analyse various microwave passive devices along with the microwave diodes by evaluating their s-parameters.
CO5	Design various active devices such as amplifiers and PLL by analysing and evaluating their performance parameters.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	-	1	-	-	-	-	3
CO2	3	3	3	3	3	-	-	-	3	3	-	3
CO3	3	3	3	3	3	-	-	-	3	3	-	3
CO4	3	3	3	3	3	-	-	-	3	3	-	2
CO5	3	3	3	3	3	-	-	-	3	3	-	3

Recommended Study Material:

❖ Text Books:

1. Microwave Engineering, Annapurna Das, 2nd edition. Tata McGraw Hill
2. Microwave devices and circuits, by Samuel Liao, PHI
3. RF Microelectronics, Behzad Razavi, 2nd Edition, Pearson Publication

❖ Reference books:

1. Microwave Technology, Dennis Roddy, PHI
2. Electromagnetic waves, Prof. R K Shevgaonkar, Tata McGraw Hill
3. Microwave Engineering, D.M Pozar, 2nd edition, Wiley Publication.

❖ Simulators:

1. High Frequency Structure Simulator (HFSS)
2. CST Studio Suite

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC453: WIRELESS COMMUNICATION

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Broadband Wireless Channel Modelling	10
2.	Diversity	10
3.	Introduction to MIMO	10
4.	Long-Term-Evolution (LTE) 4G Cellular Networks	10
5.	5G technology	05
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

- | | | |
|---|-----------------|------------|
| 1. Broadband Wireless Channel Modeling | 10 Hours | 15% |
| 1.1 Wireless channel and fading | | |
| 1.2 Raleigh fading and BER for wired communication | | |
| 1.3 Raleigh fading and BER for wireless communication | | |
| 1.4 Wireless Channel and Delay Spread | | |
| 1.5 Coherence Bandwidth of the Wireless Channel | | |
| 1.6 ISI and Doppler in Wireless Communications | | |
| 1.7 Doppler Spectrum and Jakes Model | | |

2	Diversity	10 Hours	25%
2.1	Introduction to diversity		
2.2	Transmit diversity, MIMO beamforming		
2.3	Receive diversity		
2.4	Maximal Ratio Combiner		
2.5	BER with Diversity		
2.6	Spatial Diversity and Diversity Order		
3	Introduction to MIMO	10 Hours	20%
3.1	MIMO system model		
3.2	Introduction to SVD and SVD based Optimal MIMO transmission capacity		
3.3	OSTBC and introduction to VBLAST Receiver		
3.4	Linear and Non Linear Receiver : ZF, MMSE		
3.5	ZF VBLAST, MMSE VBLAST		
3.6	MAP and ML Receiver		
4	Long-Term-Evolution (LTE) 4G Cellular Networks	10 Hours	20%
4.1	Network architecture		
4.2	Physical layer		
4.3	Resource management		
4.4	Scheduling		
4.5	Security		
5	5G Technology	05 Hours	20%
5.1	Non Orthogonal Multiple Access		
5.2	mmWave		

Course Outcomes (CO)

At the end of the course student will be able to:

CO1	Understand the fundamentals of wireless communication by providing applications to wireless communication problems.
CO2	Classify the key aspects of Wireless communication systems like transmit diversity, receive diversity, MIMO technology and apply the fundamentals of wireless communication for performance analysis of error probability and evaluate the BER.
CO3	Determine mathematically various wireless communication techniques with distinct parameters like Spectral efficiency, data rate, code rate etc.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	2	1	-	-	-	-	-	1
CO2	3	3	3	3	3	-	-	2	2	-	-	2
CO3	3	3	3	3	2	-	-	2	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Digital Communications by John G Proakis McGrawHill Science/Engineering/Math.
2. Wireless Communication by Andrea Goldsmith – Cambridge University Press
3. Fundamentals of Wireless Communications by David Tse and Pramod Viswanath, Publisher Cambridge University Press.
4. Wireless Communication by Yogesh N. Trivedi, Mahajan Publication.

❖ Simulators:

1. MATLAB

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC484: FIBER OPTIC COMMUNICATION TECHNOLOGY

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Overview of Optical fiber Communications	02
2.	Optical fibers : Structures, Wave guiding and fabrication	04
3.	Signal Degradation in Optical fibers	04
4.	Optical Sources	04
5.	Power Launching and Coupling	04
6.	Photodetectors	05
7.	Optical Receiver Operation	06
8.	Transmission Systems	06
9.	WDM Concepts and Components	05
10.	Advances in Optical Fiber Systems	05
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

1. **Overview of Optical fiber Communications** **2 Hours 3%**
 - 1.1 Electromagnetic spectrum Evolution of fiber optic system

1.2	Elements of an optical fiber transmission link		
2.	Optical fibers : Structures, Wave guiding and fabrication	4 Hours	10%
2.1	Optical laws and definitions, optical fiber modes and configurations, Mode theory, single mode fibers, step index fibers, and graded index fibers		
2.2	Fiber materials, fabrication and mechanical properties, fiber optic cables		
3.	Signal Degradation in Optical fibers	4 Hours	11%
3.1	Attenuation, signal distortion in optical waveguides		
3.2	Pulse broadening in graded index fiber, mode coupling		
4.	Optical Sources	4 Hours	10%
4.1	Light emitting diode (LEDs)-structures, materials, Figure of merits, characteristics & Modulation		
4.2	Laser Diodes -Modes & threshold conditions, resonant frequencies, structures, characteristics and figure of merits		
4.3	Single mode lasers, Modulation of laser diodes, temperature effects. Light source linearity		
5.	Power Launching and Coupling	4 Hours	11%
5.1	Source-to fiber power launching, Lensing schemes, fiber-to-fiber joints		
5.2	LED coupling to single mode fibers, fiber splicing, connectors		
6.	Photodetectors	5 Hours	10%
6.1	Principles of operation, types, characteristics		
6.2	Figure of merits of detectors photodiode materials		
7.	Optical Receiver And Transmission System	6 Hours	11%
7.1	Receiver operation, Preamplifier types		
7.2	Specification of receivers		
8	Transmission Systems	6 Hours	11%
8.1	Point –to-point link –system requirements and design of link		
8.2	Multichannel Transmission Techniques		
9	WDM Concepts and Components	5 Hours	11%
9.1	Principle of WDM		
9.2	Passive optical components		
9.3	Tunable sources and filters		
10	Advances in Optical Fiber Systems	5 Hours	12%
10.1	Telecommunications & broadband application		
10.2	SONET/SDH		

Course Outcomes (CO):

At the end of the course student will be able to :

CO1	Understand the fundamental of optical communication and structure of fiber cable.
CO2	Apply sources of attenuation, distortion and dispersion of optical signal.
CO3	Analyze optical sources and detector in optical communication and design of optical link.
CO4	Evaluate the power budget in optical cable and coupling techniques of optical fiber.
CO5	Investigate about advancement in optical communication and its applications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	2	-	-	-	-	-	-
CO2	2	3	-	3	3	-	-	-	-	-	-	-
CO3	3	2	2	3	2	1	-	-	-	-	-	-
CO4	2	3	-	3	1	-	-	-	-	-	-	-
CO5	3	-	-	2	3	-	-	-	-	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Text Book:

1. Gerd Keiser, Optical Fiber Communication, McGraw Hill Publication, 3rd Edition

❖ Reference books:

1. John M. Senior, Optical Fiber Communication
2. Djatar Mymbaev & Lowell L Scheiner, Fiber optical communication Technology
3. Zanger & Zanger, Fiber Optical Communication & Application, Mc Milan Publications

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC486: MINOR PROJECT

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	100	100	

Outline of the Course:

The Project should cover specifications, analysis, design, modelling, simulations and implementations of related work and finally it will be supported by test and measurements. Candidate will be required to complete the work within 06 months after the registration of 7th semester. After successful completions of project work, final report should prepare and submitted to the department.

Total Hours (Lab) : 30

Total Hours (Theory): 0

Total Hours:30

Detailed Syllabus:

The student can carry the project work individually or by group of 3 students. The student will be given choice to select the project from areas of Electronics & Communication engineering after discussions with instructor. The topic should be related to the field of electronics and Communication engineering. Topic can be (1) Antenna based projects (2) VLSI based projects (3) Embedded system based projects

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Understand algorithm/circuit/configuration for the project.
-----	---

CO2	Apply implementation method, list components or parts required.
CO3	Analyse Outline practical difficulties faced during implementations and device to solve them
CO4	Evaluate results of projects with other similar design specifications
CO5	Investigate Technical Document which defines the designed project details.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	-	-	3	-	-	-
CO2	3	3	3	3	3	3	-	-	3	-	3	-
CO3	3	2	2	3	3	2	-	-	3	-	2	-
CO4	3	-	-	-	-	-	-	-	3	-	2	3
CO5	-	-	-	-	-	-	-	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Instructional Method and Pedagogy:

The Course instructor will follow instruction methodology and pedagogy as follows:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Attendance is compulsory in laboratory which carries a 10% component of the overall evaluation.
- In each subsequent lab student have to regularly update about progress of the Group project. The student/group should maintain a logbook of activities. Work done, problem faced and solution evolved can be reported in logbook.
- It is preferable to publish at least one paper in national conference.
- Student/Group has to submit a project report. Evaluation will be made according to progress of the project.

Student Learning Outcomes:

After successfully completing this course, the student shall be able to:

- Understand, plan and execute a Group Project with team.
- Developing simulation model and/or implementation of model.
- Prepare a technical report based on the group project.
- Deliver technical seminar based on the Group Project work carried out.

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC456: SUMMER INTERNSHIP-II

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	3	-	3
Marks	-	100	-	

Pre-requisite courses:

Programming Language, Software Engineering.

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Students have to submit Final Project Report, Project Setup file with Source code and Project Presentation (PPT) at the time of final submission

Course Outcomes (CO):

At the end of the course student will be able to

CO1	Summarize theory and practical concepts.
CO2	Assess interests and abilities In the field of study.
CO3	Perceive career alternatives prior to graduation.
CO4	Build a record of work experience.

CO5	Adapt the skill of Team work.
-----	-------------------------------

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3	-	-	-	2	-	2	-
CO2	3	3	3	3	3	3	-	3	3	3	3	3
CO3	3	3	3	3	3	3	-	3	3	3	3	3
CO4	3	3	3	3	3	-	-	-	2	-	2	-
CO5	3	3	3	3	3	3	-	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. Books, Magazines ,Journals & online course platforms of related topics

❖ Web Materials/ Reading Material:

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. <https://www.udemy.com/>
5. <https://www.udacity.com/>
6. <https://nptel.ac.in/course.html>
7. <https://www.futurelearn.com/>

SOFTWARES:

Xillinx Altera
MATLAB/PYTHON
HFSS

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC481: DIGITAL IMAGE PROCESSING

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction	02
2.	Digital Image Fundamentals	04
3.	Image Enhancement In The Spatial Domain	06
4.	Image Enhancement In The Frequency Domain	07
5.	Image Restoration	07
6.	Introduction to Color Image Processing	06
7	Morphological Image Processing	06
8	Image segmentation	07
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

- | | | | |
|-----------|---|----------------|------------|
| 1. | Introduction | 02Hours | 07% |
| 1.1 | The Origins Of Digital Image Processing | | |
| 1.2 | Example Of Fields that Use Digital Image Processing | | |
| 1.3 | Fundamental Steps in Digital Image Processing | | |

1.4	Components of an Image Processing System		
2	Digital Image Fundamentals	04Hours	10%
2.1	Elements of Visual Perception		
2.2	Light and the Electromagnetic Spectrum		
2.3	Image Sensing and Acquisition		
2.4	Image Sampling and Quantization		
2.5	Some Basic Relationships between Pixels		
2.6	Linear and Nonlinear Operations		
3	Image Enhancement in The Spatial Domain	06 Hours	18%
3.1	Background		
3.2	Some Basic Gray Level Transformations		
3.3	Histogram Processing		
3.4	Enhancement Using Arithmetic/Logic Operations		
3.5	Basics of Spatial Filtering		
3.6	Smoothing Spatial Filters		
3.7	Sharpening Spatial Filters		
3.8	Combining Spatial Enhancement Methods		
4	Image Enhancement in the Frequency Domain	07 Hours	13%
4.1	Introduction to the 2D Fourier transform		
4.2	Smoothing frequency-domain filters		
4.3	Sharpening frequency domain filters		
4.4	Homomorphic filtering		
4.5	Implementation		
5	Image Restoration	07 Hours	13%
5.1	A model of the image degradation/Restoration Process		
5.2	Noise Models		
5.3	Restoration in the process of noise only-spatial filtering		
5.4	Periodic Noise Reduction by Frequency domain filtering		
5.5	Linear Position invariant degradation and Estimation		
5.6	Inverse filtering and Minimum mean square error (Wiener) filtering		
5.7	Constrained Least square filtering		

5.8	Geometric Mean filtering		
5.9	Geometric transformation		
6	Introduction to Color Image Processing	06Hours	13%
6.1	Color Models and its Fundamentals		
6.2	Pseudocolor Image Processing		
6.3	Basics of Full-color Image Processing		
6.4	Color Transformation		
6.5	Smoothing and Sharpening		
6.6	Noise in Color Images		
7	Morphological Image Processing	06Hours	13%
7.1	Introduction and preliminaries		
7.2	Dilation and Erosion		
7.3	Opening and closing		
7.4	The hit or miss transformation		
7.5	Basic morphological operation		
8	Image Segmentation	07Hours	13%
8.1	Detection of discontinuity		
8.2	Edge linking and boundary detection		
8.3	Thresholding		
8.4	Region based segmentation		

Course Outcomes (CO)

At the end of the course student will be able to:

CO1	Understand the need for image processing for various applications
CO2	Apply Image enhancement techniques for in spatial and frequency domain
CO3	Analyze the causes of degradation in image capturing process and to use restoration techniques
CO4	Evaluate the meaningful information from the set of images using morphological processing and Image segmentation algorithms
CO5	Investigate process of color images.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	2	-	-	-	-	-	1
CO2	2	3	-	1	3	1	-	1	-	2	-	-
CO3	2	3	-	2	3	-	2	-	-	-	1	-
CO4	3	3	-	3	3	-	-	-	2	-	-	-
CO5	3	2	-	3	3	-	-	-	-	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Text Books:

- i. Gonzalez & Wood, “Digital Image Processing”, 3rd Edition.
- ii. A.K.Jain, [Fundamentals of digital image processing](#), PHI Publication

❖ Reference Books:

1. Sanjay Sharma, “Digital Image Processing “ S K Katariya & Sons
2. S Jayaraman, “Digital Image Processing ” , 1st Edition, Tata Mcgraw Hill

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC482: RADAR SYSTEMS

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Principles of Radar	06
2.	MTI & Pulse Doppler Radar	07
3.	CW and FMCW Radar	08
4.	Navigation and Tracking Radars	08
5.	Direction Findings	08
6.	Radio Ranges and Landing Systems	08
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

- | | |
|--|---------------------|
| 1 Principles of Radar | 06 Hours 15% |
| 1.1 Introduction Radar frequencies, | |
| 1.2 Radar Equation, Radar Block Diagram | |
| 1.3 Radar Applications, | |
| 1.4 Receiver Noise, Various Displays | |
| 1.5 Pulse Repetition Frequency, Pulse Duration | |

2	MTI & Pulse Doppler Radar	07 Hours	20%
2.1	Introduction, Operation of MTI Radar,		
2.2	MTI receiver with delay line canceller		
2.3	Multiple or staggered, Range gated Doppler filters		
2.4	MTI from a moving platform		
2.5	Introduction to pulse Doppler Radar		
3	CW and FMCW Radar	08 Hours	15%
3.1	CW Radar		
3.2	FMCW Principle, Block diagram of FMCW radar,		
3.3	Radio altimeter		
4	Navigation and Tracking Radars	08 Hours	15%
4.1	Four Methods of Navigation.		
4.2	Airport Radar, Meteorological Radars,		
4.3	Navy Radar, Remote Sensing Radar,		
4.4	Synthetic aperture Radar, Side looking Airborne Radar(SLAR)		
5	Direction Findings	08 Hours	20%
5.1	Introduction, Loop Antenna, Sense Finder		
5.2	Increasing the Sensitivity of Direction Finder		
5.3	Goniometer method of increasing Loop Voltage		
5.4	Errors in Direction Finding, Automatic Direction Finder		
6	Radio Ranges and Landing Systems	08 Hours	15%
6.1	Four course Radio range		
6.2	VOR, Doppler VOR,		
6.3	DME, TACAN, Ground controlled Approach		
6.4	ILS, MLS		
6.5	Hyperbolic Navigation, LORAN, DECCA		

Course Outcomes (CO):

At the end of the course student will be able to:

CO1	Undersatnd fundamentals of Radars.
CO2	Analyze and examine the working principle of MTI, Pulse Doppler Radar and FMCW Radar.

CO3	Evaluate fundamentals of direction finding in radar systems through various methods and mathematical equations.
CO4	Investigate theory of various types of radar and navigation systems.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	3	-	1	-	3	3	-	3
CO2	2	3	3	-	-	3	-	-	3	3	-	3
CO3	3	2	-	3	-	-	-	2	-	2	3	3
CO4	3	3	-	-	2	-	2	-	1	2	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Text Books:

1. Introduction to Radar System by Skolnik (TMH)
2. Radar Systems and Radio AIDS to Navigation by Dr. A.K. Sen& A.B. Bhattacharya ‘Khanna Pub’
3. Elements to Electronic Navigation by Nagraj (TMH)

❖ Reference Books:

4. Radar Principles by Peyton Z. Peebles, Johnwiley, 2004
5. Principles of Radar by J.C Toomay, 2nd Edition –PHI, 2004

❖ Web material:

1. <http://www.radartutorial.eu/index.en.html>
2. <http://www.vectorsite.net/ttradar.html>

http://www.tpub.com/content/neets/14190/css/14190_13.htm

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC483: ERROR CONTROL CODING

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Error Control Coding	08
2.	Coding in Wireless Digital Communication	15
3.	Modern Iterative Coding	22
	Total Hours (Theory):	45
	Total Hours (Lab):	30
	Total Hours:	75

Detailed Syllabus:

1	Introduction to Error Control Coding	03 Hours	10 %
1.1	Introduction to traditional and modern coding theory		
2	Coding Over Finite Field	20 Hours	40 %
2.1	BCH code, Construction of BCH code for given minimum distance, Vandermonde matrices, BCH bound.		
2.2	Properties of BCH codes, their representation as polynomials, minimum polynomials.		
2.3	Minimum polynomials, their construction and properties, their connection with cyclic codes, Generator polynomial Of a cyclic code.		

2.4	Dimension of BCH codes, Examples of BCH codes, Systematic encoding, Syndrome decoding for BCH codes, Error Locators.		
2.5	Reed-Solomon (RS) Codes, Dimension, Definition of distance, weight in $GF(2^m)$, Generator polynomial, Minimum distance and binary expansion of RS codes		
2.6	Reed-Solomon (RS) Codes: Decoding overview, PGZ Decoder for RS codes.		
2.7	Reed-Solomon codes in practice: erasure decoding, burst erasure correction, some modern decoders.		
3	Modern Iterative Coding I	20 Hours	40 %
3.1	Turbo Codes : Encoders, Interleavers, Turbo Decoders		
3.2	Low Density Parity Check Codes		
3.3	Socket construction of regular LDPC codes, Tanner Graphs, Neighbourhoods and cycles in graphs.		
3.4	Gallager A decoding algorithm for LDPC codes and its analysis, LDPC Threshold.		
3.5	LDPC Code in 5G, Encoding LDPC Code in 5G		
3.6	Introduction to TCM (Trellis Coded Modulation)		
4	Application of Error Control Coding	02 Hours	10 %

Course Outcomes (CO)

At the end of the course student will be able to:

CO1	To understand the fundamental concept and need of mathematical theory of communication for various applications.
CO2	To understand the basic principles of information theory to achieve error free communication over noise channel, uniquely decodable code for source encoding and Is there any uniquely decodable code is available for given length.
CO3	To realize the various type of channels and receiver data detection algorithm available in error control coding and use of appropriate detection algorithm in real time application.
CO4	A valuation for present day Capacity moving toward codes like Turbo and LDPC codes, their encoding and deciphering systems and execution assessment.
CO5	After culmination of this course, understudies ought to have the option to Work as fashioners of divert codes in Physical Layer structure and Storage framework configuration groups in industry and Work in the scholarly community/industry as specialists blending new channel codes for new applications in Wireless/Wired correspondence frameworks and Storage frameworks.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	3	-	3	-	-	1	-	1	-	-
CO2	3	-	3	3	3	-	-	-	2	1	-	3
CO3	3	3	3	3	3			2	2			2
CO4	3	3	3	3	3			2	2			2
CO5	3	3	3	3	3			2	2			2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

Recommended Study Material:

❖ Reference Books:

1. B. P. Lathi, Modern Digital and Analog Communication Systems, Oxford Press.
2. Thomas M. Cover and Joy A. Thomas, Elements of Information Theory, John Wiley
3. Shu Lin, Daniel J, Costello, Error Control Coding, Pearson Education

❖ Simulators:

3. MATLAB

**B. Tech. (Electronics and Communication Engineering)
Programme**

SYLLABI
(Semester – 8)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
V.T. PATEL DEPARTMENT OF ELECTRONICS AND COMMUNICATION
EC458: PROJECT

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	36	36	18
Marks	-	600	600	

A. Detailed Syllabus:

As per the objective and problem area chosen.

B. Instructional Methods and Pedagogy:

C. Student Learning Outcomes / objectives:

- Able to understand the basic importance of advanced phase of the mini project
- Able to present him/herself in front of the public i.e killing the stage fear and building the self-confidence to deal with real time challenges even in emergencies like breakdown etc.
- Able to study and work at his/her own in the area of interest.
- Develop the soft skill and personality
- Able to learn multiple new techniques
- Able to know the current trends in the electronics and communication engineering
- To get exposure for other integrated areas such as market survey for technical components, cost effectiveness for purchasing them.

D. Recommended Study Material:

- ❖ Reference books:
- Depends upon the area of project problem chosen and finalized

by the respective faculty advisor

❖ Web materials with full citations:

- Depends upon the area / topic / problem of project chosen and finalized by the respective faculty advisor.

❖ Supporting materials

- Printed Books, Magazines, Journals
- . • Software
- . • Hand Outs
- . • E-resources such as web sites, on-line magazines and journals